

THESIS

BENNOUR MOHAMED HSIN

2025-12-03

Table of contents

Preface	6
1 Introduction	7
1.1 General background	7
1.2 Problem statement	10
1.2.1 Relatedness, relatedness density, and diversification	10
1.2.2 Empirical consequences	11
1.2.3 Research problem	13
1.3 Objective	17
1.4 Research questions and hypothesis	17
1.5 Structure	18
2 Literature review	20
2.1 Regional diversification as an outcome to resilience: Overview from an Evolutionary perspective	20
2.2 Why place (still) matters?	23
2.3 Relatedness as the baseline, context as the extension	31
3 The interconnectedness between regions and technologies	40
3.1 Methodological Motivation	40
3.2 The mainstream approach	44
3.2.1 Relatedness Through Co-occurrence	44
3.2.2 Structural Limitations of Co-occurrence-Based Relatedness	47
3.2.3 Relatedness Density Through Linear Aggregation	49
3.2.4 Structural Limitations of Linear Aggregation	50
3.2.5 Implications for Contextualisation	52
3.3 An Alternative Approach: Machine Learning for Contextualised Relatedness	54
3.3.1 Machine Learning as Prediction Framework	54
3.3.2 Why Random Forest	56
3.3.3 Random Forest Algorithm	58
3.3.4 Training Procedure	61
3.4 Technological Potential	62
3.5 Feature Importance Technology Space (FITS)	63
3.6 Coherence: Bridging Technology Networks and Regional Portfolios	69
3.6.1 Why we introduce coherence here (and why at IPC-section scale)	70
3.7 Summary	73

4	Empirical Design	75
4.1	Overview and Identification Goals	76
4.2	Research Desgin	78
4.3	Data	82
4.3.1	Outcome: Regional Technological Potential	83
4.3.2	Technology–Region Level (Dyadic) Variables	84
4.3.3	Technology Level Variables	86
4.3.4	National Level Variables: GEI Pillars	87
4.3.5	Region Level Variables (Spatial Models)	90
4.4	Model 1: Dyadic Fixed Effects Models	92
4.5	Model 2: Hierarchical Models	93
4.6	Model 3: Spatial Panel Models	95
4.6.1	Spatial Autoregressive (SAR) Panel - Primary Specification	97
4.6.2	Spatial Error Model (SEM) - Robustness	98
5	Results	99
5.1	Entry	100
5.2	Network Effects and Coherence Moderation	101
5.3	National Context and Robustness	105
5.4	Hierarchical Structure and Spatial Implications	106
5.5	Role of space	109
6	Summary	115
6.1	Discussion	115
6.1.1	On the methodology	116
6.1.2	On the results	117
6.1.3	Policy implication	118
6.2	Future research agenda	118
6.3	Conclusion	118
	References	119

List of Figures

3.1	Technology Network (based on 2018 patent data)	65
4.1		84
4.2		85
5.1		113

List of Tables

4.1		87
4.2		88
4.3		89
4.4		89
4.5		90
4.6		91
4.7		91
4.8		96
5.1		102
5.2	logit(pred1) - ols	104
5.3	logit(pred1) - ols	107
5.4	logit(pred1) - ols	110
5.5		111
5.6		114

Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

1 Introduction

1.1 General background

The Covid 19 pandemic showed structural challenges of national economies all over the world, specifically the fragility of neoliberal policies in times of crisis and the lack of industrial and economic resilience. Six years after the fact, our societies are now confronted with inevitable novel challenges and looming shocks. We are already witnessing the consequences of AI development as it paves the way for a new technological revolution that would render most local economies obsolete and cause massive unemployment in white collar sectors. Sadly, that doesn't seem to be the end of the shocks the world has seen recently as the war in Ukraine, Gaza, Iran, and most importantly the current, chaotic trade wars, seem to foster ever increasing uncertainties. Facing all this, policy makers are confronted with a simple choice; strategise and plan for more resilient local economies. In different streams of literature, resilience is directly related to diversification/variety, whether in portfolio management in finance, trade partnerships/linkages, industrial activities, or in terms of knowledge as well. Thus we can equivalently say that resilience is the capacity

to resist and/or adapt to external shocks by relying on exiting internal capabilities that evolve in the face of such shocks. This means that for an economy to survive uncertainties, it needs to evolve, change, and innovate its way to the other end of the bleak challenges it's confronted with. However, to evolve, change and innovate, a baseline of knowledge should be leveraged since the consensus is that variety is a buffer against external shocks and shields uncertainties. This chain of thoughts, takes us back to the conceptual basics of sustainable economic development and growth; the knowledge fabric is what facilitate any long term strategy, and has been shown in the literature as clear catalyst for societal prosperity and economic resilience. For this reason the study of knowledge is detrimental for policy making, and understanding how to increase its diversity is more relevant than ever before.

Knowledge is a set of information that covers one or many topics, and its characteristics are contingent on the different forms it can take or how it was created, generally speaking, academia and businesses are the main knowledge creators in any economy through research and patents. Essentially, knowledge can be codified (accessible by anyone through any medium), or tacit (personal information based on social connections, intuition, experience, etc, that's hard to share with others). The consensus in the literature is that the main driver of competitive advantage for firms is the tacit form of knowledge, which is also widely acknowledged that it's space dependent. However, the knowledge produced by firms can be reliably seen in patents, although they capture codified information, they also reveal tacit knowledge and its geographic footprint in space. This means that its detrimental to assess the knowledge in patenting activities (we refer to this knowledge as

technologies), and more so to focus on the local aspect of these activities i.e: sub-national regions.

This framing, however, is not at all new or novel. In fact, this is the entire aim of the literature of the geography of innovation; to study how innovation is created and diffused to different actors in different geographical contexts. Specifically, relatedness and economic complexity (REC) is one of the main streams of literature that focus on the relationships between activities and geographies. The conceptual and methodological framework that REC provides is widely used and adopted in academia and among policy practitioners and was one of the main contributors to the smart specialisation policy literature. The ideas embedded in this framework, to put it simply, rely on the premise of spatial dependence of tacit knowledge in local/regional economies/geographies and focus on simplifying these relationships using network science to model the relationships between knowledge and regions. Albeit these simplifications provide valuable scope for analysis and interpretation, the cost from the loss of granular information implies that there's much more conceptual, methodological, and empirical work needed. The reason for this is because the loss of information bias the empirical interpretation in the sense that we end up with a homogeneous implication with weak regards to the regional and national contexts as well as the technological characteristics. This work is motivated by this gap and aims to simply contextualise the study of knowledge diversification using the same granular information publicly available and commonly used in the REC literature. The idea is simple, account for endogenous and exogenous contexts using granular data and understand the contexts and contingencies that drives regional diversification.

1.2 Problem statement

The main problem that this work aims to solve is directly embedded in the methodological and empirical framework of REC. REC models diversification through network aggregation based on co-location and co-occurrence patterns. Using these patterns, different aggregations are used to quantify relatedness (the frequency of observing a pair of activities in the same region), and relatedness density (how much of the activities frequently observed together a region has). However, these measures are often interpreted not as aggregations of frequent observations but rather as relationship models. Empirically, diversification is studied using these constructs as predictors of entry, that is a region's entry to a new specialisation in a new technology, often considered as a binary outcome. In here we briefly outline the big picture in the REC methodological and empirical framework, its conceptual issues, its empirical consequences, and highlight the research gap.

1.2.1 Relatedness, relatedness density, and diversification

Relatedness and relatedness density are essentially measures of proximity. In a sense they describe how close two technologies are close to each other, or how close a given technology is to a region given its portfolio of technologies. To further decompose the problem here, we will first establish the methodological constructs for proximity measures. For relatedness that's co-occurrence, and for relatedness density that's the linear aggregation of relatedness.

First, co-occurrence is essentially the frequency of observing two activities together. In

the REC literature, this frequency describes the strength of the relationship. Activities frequently observed together are more related than the pairs rarely observed together.

Second, linear aggregation of relatedness essentially measures the percentage of co-located technologies in a region that are related to a reference technology. Thus, we can think of relatedness density as the link between related technologies and co-located technologies. The idea in the REC literature states that relative to a given technology, the more related technologies a region has, the more likely that it can develop that technology.

These two constructs are used together to predict the probability that a region will enter a new technology. The REC literature shows that relatedness density is consistently associated with higher probabilities in almost all studies. These results among others were one of the major latent contributors to smart specialisation strategy (S3) policy. Thus the consensus in the literature was clear: In order to diversify into new technologies with the highest likelihood of success, regions must prioritise investment in related technologies.

1.2.2 Empirical consequences

The idea of resilience is not a main focus for the REC literature nor it is ours. However, falling back to this concept allows us to further assess the empirical consequences of the mainstream interpretation of relatedness and relatedness density. The idea is that in order to be resilient to external shocks and subsequent uncertainties diversification is key. But what kind of diversification is required and feasible and how to achieve it is the focus here. The REC literature tells us that the most likely successful diversification strategy

is the one that targets related capacity in the regions, often referred to as related variety. However, generalising this recommendation is not that straight forward. Aggregate regional capacities and their national and broader geographic contexts differ significantly. The initial landscape of the regional technological portfolio is detrimental here because this strategy could favour regions with already diverse portfolio but it's questionable that regions with limited portfolios would equally benefit. This is aligned with the concept of path dependency, related variety without context enforces that dependency and locks regions within their limited capacities. This brings us back to the core focus of in this work; context is key. However, context on its own here might not be enough since path dependency of related variety is a direct result of how relatedness and relatedness density is calculated and interpreted. The constructs that enable these measures (co-occurrence and linear aggregation) are the core problem that we're highlighting here. The reason behind this specific focus relied on the implicit assumptions embedded in these methodological constructs.

Co-occurrence assumes that technologies frequently observed together are likely related. Although this is within the boundaries of common sense, it's highly unlikely that's actually the case. A frequency measure is only informative when we have more observations than items—that is, more geographies than technologies. Almost always, we will have more technologies than geographies, this means that relatedness is at best noisy. Additionally, relatedness as interpreted in the literature quantifies the relationship between pairs of technologies. However, as it stands, there's no differentiation in the direction of that relationship thus, assuming that the relationship between two technologies is sym-

metrical. Albeit this assumption in itself is not problematic, it exacerbates the linearity issue when we measure relatedness density as we lose information in co-occurrence, symmetry, and linear aggregations. This takes us to the final issue we would like to highlight; relatedness density. Simply put, relatedness density measures the sum of technologies related to a reference technology present in a given region. The implicit assumption in here is that technologies are linked through linear combinations, and those combinations predict the likelihood of successful diversification. However, relatedness density is often interpreted as a value that quantifies the existing requirements a region has relative to a technology, whereas the sum of existing related technologies do not inform us on the actual requirements.

In summary, relatedness and relatedness density measures suffer from diverse methodological issues embedded in the implicit assumptions in their core constructs. Co-occurrence and linear aggregation of observed frequency are misinterpreted, accrues information loss, and poorly handles the granular data often used. This means that the empirical and methodological work ahead must account for these issues to further contextualise the study of diversification strategies.

1.2.3 Research problem

In the light of all the mentioned in this section, we fall back again on the core idea that we started this text with; How can we contextualise diversification strategies? The answer to this question is multi-layered and complex. In this section we started by outlining the importance of diversification strategies for regions into new technologies via patenting

activities. We explained that the REC literature provides interesting methodological and conceptual framework of analysis and showed that despite their usefulness they suffer from structural issues that limit the advantages of the used granular data, thus limit the incorporation of a broader context empirically. Essentially, the research problem we focus on here, is both methodological and empirical in nature. We highlighted the structural methodological issues as the research gap which will be the focus of our methodological and empirical contribution.

To address the methodological gaps in REC, we propose a multilayered contextual framework that distinguishes between different contextual factors. Relatedness measures alone conflate these dimensions. By decomposing context into endogenous technology-region fit and exogenous environmental conditions, we can identify which combinations enable successful diversification across different regional contexts, following this design:

Endogenous factors (within-region characteristics):

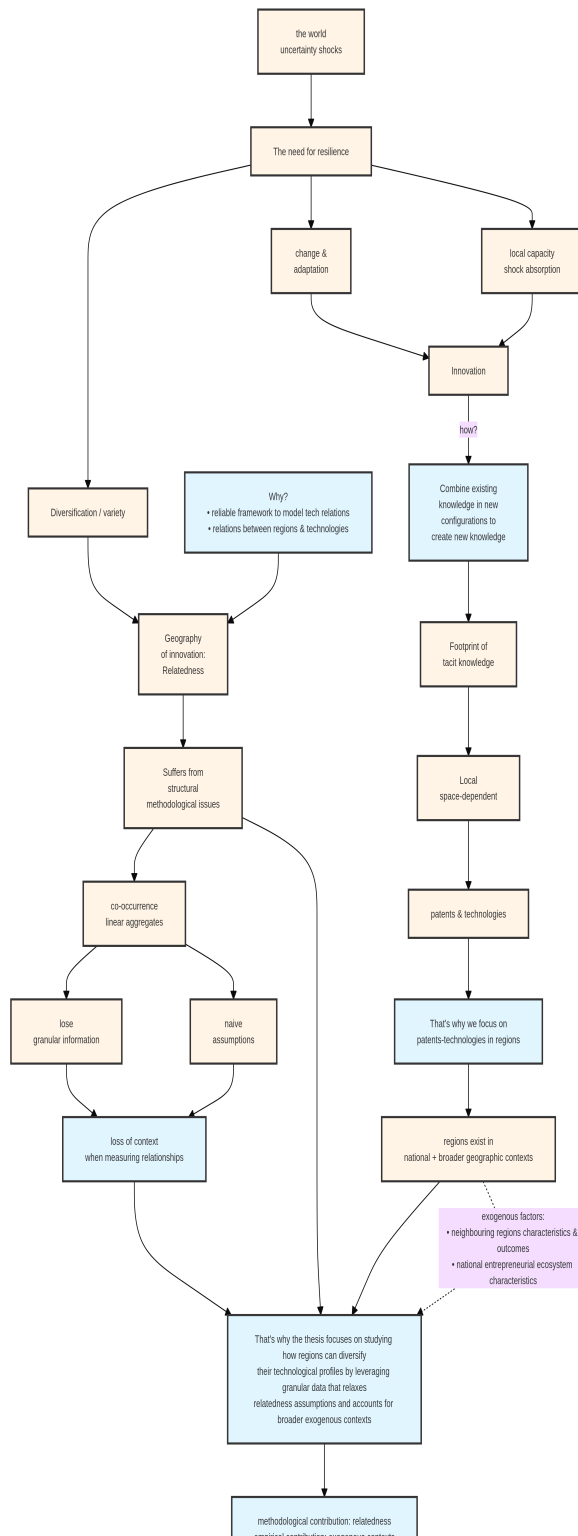
- Technology-specific attributes (network position, centrality, embeddedness)
- Regional knowledge infrastructure (coherence between technology relationships and regional specialization patterns)
- Regional technological volume (number of patents in a specific technology)

Exogenous factors (beyond-region influences):

- National entrepreneurial ecosystem characteristics (institutions, financing, R&D culture)
- Spatial spillovers from neighbouring regions (outcome diffusion, knowledge flows)

- Geographic proximity to coherent knowledge clusters

The interaction between these layers is critical: endogenous capacity determines what a region can develop, while exogenous context shapes realisation possibilities. For instance, a technology may align with regional knowledge infrastructure (endogenous coherence), but without supportive national institutions or proximity to innovative neighbours (exogenous support), entry probability remains low.



1.3 Objective

The objective from this work is to extend the methodological framework of relatedness starting from the structural issues it suffers from. In order to show how our methodological contribution benefit the study of regional technological diversification we rely on an empirical study that shows how broader context can be included and how such context inform granular policy insights. The broader context we aim to include empirically relied on relaxing the implicit relatedness assumptions, explicitly include the endogenous characteristics of the technologies and regions contingent on the regional knowledge infrastructure while simultaneously account for the characteristics of the national ecosystems. With such an approach we end up accounting for more contextual layers than the mainstream approach.

1.4 Research questions and hypothesis

RQ1: How to further contextualise diversification strategy based on the relatedness framework?

- H1a: Asymmetric relatedness measures better predict technology entry than symmetric measures.

RQ2: Is successful diversification contingent on regional knowledge infrastructure?

- H2a: Technology-specific characteristics impact on diversification is contingent on the regional knowledge coherence.
- H2b: Technology-specific characteristics impact on diversification is contingent on the regional knowledge stock.

RQ3: Does the national ecosystem influence diversification?

- H3a: National Entrepreneurial Ecosystem characteristics positively affects regional entry into new technologies.

RQ4: What role does space have? Do neighbouring regions influence diversification?

- H4a: Spatial spillovers of outcome from neighbouring regions positively affect technology entry.
- H4b: Neighbouring regions' technological characteristics influence focal region diversification outcomes.
- H4c: Geographic proximity to regions with coherent knowledge infrastructure increases entry probability.

1.5 Structure

We structure this work around our methodological and empirical contributions. Given the complex nature of the problems we aim to solve, we will first start with more literature context that underlines geography of innovation, and more importantly relatedness

and economic complexity literatures. We then outline our approach in the methodology chapter where we discuss in more details measures of relatedness and proximity and provide alternative conceptualisation to modeling the relationships between technologies and regions. In the same chapter we also detail other elements that will be relevant to the empirical part specifically knowledge coherence. In the empirical chapter we outline the research design and the modeling procedures that implements our methodological contribution. The results chapter outline the results of our work and its empirical consequences and interpretation, which we discuss in more detail in the discussion and conclusion of this work in which we also outline different further extensions and future research directions.

2 Literature review

2.1 Regional diversification as an outcome to resilience:

Overview from an Evolutionary perspective

In the previous chapter, we framed this work’s core idea around the need to study technological diversification patterns of regions within a broader context. To justify this framing we grounded this core idea on the ever increasing importance of resilience facing an increasingly uncertain global and geopolitical world order (Xiao, Boschma, and Andersson 2018). Although resilience in itself is not a core focus of this work, we will use it as an adjacent conceptual paradigm to justify our core ideas conceptually, methodologically and empirically through the literature.

Within the literature of evolutionary economic geography (EEG), P.-A. Balland, Rigby, and Boschma (2015) summarize this turn toward resilience and highlight that, while interest has grown, “the resulting debate suggests [that economic geographers] are far from a common understanding”. This is to say that resilience as a concept is attracting more

attention, however, there’s still no consensus on a generalised ontological framework that drives further research in this area. The different studies that addressed this concept (eg: Christopherson, Michie, and Tyler (2010); Fingleton, Garretsen, and Martin (2012); Martin (2012); Pendall, Foster, and Cowell (2010); Pike, Dawley, and Tomaney (2010); Simmie and Martin (2010)) relatively converge in refuting equilibrium based approaches that consider resilience as an immediate reaction to shocks, and calling for further focus on an evolutionary approach. The argument for this proposed agenda is related to robust results showing that precedent regional capabilities are recycled over time with different configuration driving different regional growth paths (Boschma, Balland, and Kogler 2015; Neffke, Henning, and Boschma 2011). However, P.-A. Balland, Rigby, and Boschma (2015) also criticises the literature in this context for ignoring key explanatory dimensions that are central to evolutionary accounts of regional change; technological relatedness, network structures, and institutions.

Recent syntheses also stress that capability accumulation is a necessary basis for adaptation, but not a sufficient explanation of successful upgrading, which is conditioned by technological trajectories, shifts in demand, and the evolving capabilities of competitors (C. Pinheiro 2025; **perezsoete1988?**; **leemalerba2017?**). We adopt this triad as an organising device: relatedness and network structure describe the endogenous opportunity structure of regional capabilities, while institutions shape the exogenous enabling conditions. A useful way to make that second part operational without reducing institutions to a single rule of law proxy, is to treat entrepreneurship and institutions as a coupled system that governs how knowledge is mobilised into new ventures and new activities. This is

precisely the motivation behind the “National Systems of Entrepreneurship” (NSE) perspective, which defines entrepreneurship systems as resource-allocation systems driven by individual opportunity pursuit but regulated by country-specific institutions (Acs, Autio, and Szerb 2014).

This agenda matters for the present study because our goal is not to address regional resilience per se, but to explain the mechanisms that shape diversification feasibility in a way that is consistent with an evolutionary view of regional adaptation. Some work has been done in this context notably Tóth et al. (2022) who address the relatedness and network structures gaps. The authors anchor their view on resilience by considering it an “adaptive ability of regions to reconfigure their economic structure” and that this ability could be observed in diversification patterns. This intuition was addressed and confirmed in Rocchetta et al. (2022) for technological diversification patterns and for industrial diversification patterns in Xiao, Boschma, and Andersson (2018). However, Tóth et al. (2022) retain a density-based operationalisation of relatedness. This choice, as (tacchella2023relatednessml?) demonstrate systematically, generates measures that perform worse than trivial autocorrelation strategies in predicting genuine new specialisations—a major point we discuss further in subsequent chapters. Moreover, neither Tóth et al. (2022) nor Rocchetta et al. (2022) incorporate institutional or ecosystem conditions as contributing factors¹. This leaves open the question of whether network structure shapes feasibility unconditionally or in interaction with the enabling environment.

Against this background, we adopt an evolutionary framing that treats diversification as a

¹Xiao, Boschma, and Andersson (2018) incorporated a quality of governance measure that showed no consistent significant effect in their study

constrained, path-dependent process of recombination. Empirically, this motivates a more contextualized assessment of regional capabilities and a decomposition of diversification drivers along three dimensions that the literature highlights as consequential: relatedness, network structure, and higher-level context (including national entrepreneurial ecosystems and spatial interdependence). This framing carries particular policy urgency: as C. Pinheiro (2025) demonstrates, the dominant tools currently deployed to advise regional and national governments on diversification strategy-Relatedness and Economic Complexity (REC)-carry theoretical and empirical limitations that are consequential precisely in the kinds of catching-up or restructuring contexts where resilience is most needed. Crucially, this framing also carries methodological implications. If diversification is conditioned by portfolio-level configurations of capabilities rather than pairwise proximities, then measuring feasibility as a function of the whole capability basket-rather than as a sum of dyadic relatedness scores-is not merely a technical refinement but a theoretical necessity (tacchella2023relatednessml?; C. Pinheiro 2025). This is the foundation on which the present study’s empirical strategy rests. Understanding what shapes diversification feasibility in more granular context, therefore becomes not only an academic concern but also a critical policy question.

2.2 Why place (still) matters?

The agenda for assessing regional resilience through an evolutionary lens through relatedness, network structures and institutions, falls on the theoretical convergence that regional economies can be considered as capability systems (Hidalgo and Hausmann 2009;

Hausmann and Hidalgo 2011; Martin and Sunley 2007) where a region’s economy can be represented as a structured set of complementary capabilities that constrain which new activities are feasible. To approximate the interconnectedness between these capabilities in a regional economy, interpreting this system as a network is the consensus here (Tóth et al. 2022). In this sense, a region’s economy can be described as a capability network where nodes are technologies linked through how related they are (Boschma 2015; Whittle and Kogler 2020) and where the links represent empirical co-occurrence, co-specialization, or inferred dependence. Consequently proximity, centrality, and distance measures (among others) encode potential recombination pathways. The idea here is that such networks reflect the knowledge resources and dependencies of the region and how they are used in different configurations operationalised via a constrained search process (Napolitano et al. 2018).

From here, the direction is clear, the information contained in these networks should predict how these configuration result in different outcomes for different regions, as from this perspective place matters. Because of these place-dependent constraints of the accumulated regional knowledge structure, this search process (of new diversification opportunities) is inherently path dependent, i.e. future technological capabilities occurs near previous accumulated ones. This means that path dependency for related diversification is a feature of these network structures.

In light of this logic, (batheltstorper2023?) acknowledge the theoretical merits of these arguments, but provide a set of critical points often omitted in the literature. First, related diversification via relatedness measures are “based on a statistical artifact” and inferred

via “another artifact: the industrial [technological] classification systems”. Second, this local related variety of diversification patterns predict the local infrastructure, but do not capture the dimensions of the learning process that produced them. Third, these patterns as measured via relatedness tend to confound scale and become instead a proxy of the size of the regional economy itself. These concerns were also highlighted in Whittle and Kogler (2020) where the authors qualify them as a definitional bottleneck that confounds different mechanisms “making an all-purpose interpretation problematic”.

These critiques matter for the present study, not because they invalidate the relatedness tradition, but because they clarify what relatedness can and cannot do on its own. Measures based on observed co-occurrence in patent or industry classifications can successfully summarize regularities in diversification patterns, yet their meaning is partly constructed by data and taxonomy choices (batheltstorper2023?; Boschma 2017; Whittle and Kogler 2020). In particular, if relatedness proxies are sensitive to classification “artifacts” and scale effects, then a purely dyadic “proximity $\rightarrow$ entry” story risks conflating (i) a region’s size and measurement resolution with (ii) the underlying learning and capability-building processes that actually generate diversification (batheltstorper2023?; Boschma 2017). Additionally, from an industrial-policy perspective, this implies that relatedness-based recommendations should be treated as context sensitive rather than universal, especially in catch-up settings where historically successful upgrading sometimes required assembling capabilities beyond locally ‘related’ trajectories (C. Pinheiro 2025; amsden1989?; kimmnelson2000?; wade2004?). This motivates a shift from treating relatedness as a self-sufficient explanation toward treating it as a useful but incomplete representation of

constraints—one that must be complemented with clearer mechanisms and context.

A key step in that direction is to reintroduce agency explicitly. Regions do not diversify by themselves: “a city and its surrounding region do not act for themselves, but [...] depend on firms and entrepreneurs to introduce new activities” (Neffke et al. 2018). This agent-centered view is consistent with evolutionary accounts of change (variation, selection, retention) (nelsonwinter1982?; Martin and Sunley 2007), and it clarifies why the same relatedness structure can lead to different outcomes across places. Neffke et al. (2018) frame this using a resource-based view applied to regions, where places are endowed with capability bases that shape feasible paths of development (lawson1999?; boschma2004?; penrose1959?). While branching evidence supports relatedness-constrained recombination, novelty and structural change often arise through new establishments (especially those with nonlocal roots) rather than incumbents, implying that feasibility depends on both opportunity structure and the actors attempting entry (Neffke, Henning, and Boschma 2011; Boschma, Balland, and Kogler 2015; hidalgoklinger2007?; Neffke et al. 2018). In this view, relatedness and network position describe an opportunity structure, what is plausibly feasible given a region’s existing capabilities. Whereas higher-level context governs whether feasible options become realised diversification outcomes. This distinction aligns with recent industrial-policy discussions that treat capability proximity as informative but incomplete unless combined with institutional and market conditions (C. Pinheiro 2025; Andreoni and Chang 2019).

This perspective also sharpens the link between diversification and adaptation: beyond incremental “branching,” structural change is “unrelated regional diversification that re-

quires a transformation of the local capability base” (Neffke et al. 2018). Put differently, adaptative diversification is not only about whether a region adds new technologies, but about whether it can renew or extend its capability base to support activities that are less aligned with its inherited structure (Martin 2012).

This agent-based framing also helps reconcile “path dependence” with the empirical reality of novelty. The branching literature shows that diversification frequently occurs near what regions already do, consistent with relatedness-constrained recombination (Neffke, Henning, and Boschma 2011; Boschma, Balland, and Kogler 2015; **hidalgoklinger2007?**). Yet innovation and new paths can also emerge from recombinations that bridge previously distant knowledge domains, and from new entrants that introduce different routines and connections (Martin and Sunley 2007; **isaksentripp12014?**). Neffke et al. (2018) provide evidence for this asymmetry: incumbents tend to reinforce existing specializations, while new establishments contribute disproportionately to unrelated diversification and structural change (Neffke et al. 2018). In this sense, diversification feasibility is not only a property of regional relatedness patterns; it depends on who attempts diversification (incumbents vs entrants, local vs nonlocal actors) as well as on whether they can mobilize the resources needed to scale and on whether local environments can support the survival and scaling of novel activities (Neffke et al. 2018).

At this point, the institutional dimension becomes central in a way that remains compatible with the network framing. Institutions matter because they shape how agents access and mobilize capabilities. Neffke et al. (2018) make the mechanism explicit: access depends on embeddedness in “economic, social, and trust relations with governments,

institutions, and other firms”. This implies two channels: institutions shape access to local resources via embeddedness, and they shape reliance by affecting whether actors can substitute extra-local capabilities (e.g., through subsidiaries or external networks) (Neffke et al. 2018). This is consistent with long-standing arguments that localized institutions, norms, and relational infrastructures condition learning, coordination, and innovation in places (storper1995?; cookemorgan1998?; grabher1993?; saxenian1994?). It is also consistent with the geography of innovation view that knowledge spillovers, interactive learning, and absorptive capacity are uneven across space (cohenlevinthal1990?; audretschfeldman1996?; visserboschma2004?). The implication here is straightforward: networks describe feasible recombinations, but institutions condition whether those recombinations can be enacted and/or sustained. This is precisely why entrepreneurship and institutions are treated jointly: entrepreneurs “provide the transmission mechanism transferring advances in knowledge into economic growth”, yet that transmission “may be either hampered or facilitated by the institutional environment (Acs et al. 2018). This ecosystem perspective also aligns with our framing because it provides a disciplined way to think about which parts of the institutional environment matter for mobilisation, rather than treating institutions as a single latent factor. The measurement framework explicitly highlights three features: a systemic approach (components interact), a “Penalty for Bottleneck” logic (weakest links constrain performance), and contextualization (entrepreneurship is embedded in institutional frameworks). This is the rationale behind the Global Entrepreneurship Index (GEI) (Acs, Autio, and Szerb 2014): it is built from pillars that combine individual and institutional variables, and it operationalizes bottlenecks by penalising underperforming pillars rather than allowing them to be fully offset by strong

ones. Importantly, the GEI (and its different derivatives) is among the few ecosystem measures explicitly designed to integrate both individual and institutional dimensions, precisely because ecosystem performance is not reducible to institutions alone.

This is where the entrepreneurial ecosystem perspective becomes analytically useful for the present study: it provides a structured way to approximate the institutional and systemic conditions that make diversification feasible beyond what dyadic relatedness captures. Innovation-systems research makes the same point in functional terms: “innovation systems are a very important determinant of technological change” (Hekkert et al. 2007). Rather than treating institutions as a static backdrop, the functional approach focuses on what the system does—supporting experimentation, knowledge creation, diffusion through networks, resource mobilization, market formation, and legitimacy building (Hekkert et al. 2007). In this view, entrepreneurs are essential because they turn “the potential of new knowledge, networks and markets into concrete actions” (Hekkert et al. 2007), but their effectiveness depends on the broader system functions that provide resources and reduce resistance to new activities (Hekkert et al. 2007; **sabatier1988?**). Seen this way, ecosystem conditions are not just ‘background’: they influence whether regions can mobilise finance, legitimacy, and complementary inputs fast enough to exploit shifting markets and technological transitions. This resonates with recent REC-for-policy reviews that emphasise timing and competition as central moderators of capability-based opportunity (Hekkert et al. 2007; C. Pinheiro 2025; **perezsoete1988?**). Entrepreneurial ecosystems translate these functional ideas into a place-based lens by focusing on how finance, human capital, universities, intermediaries, culture, infrastructure, and governance

cohere locally to enable productive entrepreneurship (**stamwelter2020?**; **audretschbelitski2017?**; **brownmason2017?**). They also emphasize that ecosystems are spatially embedded but shaped by nested regional, national, and global structures, matching the idea that innovation systems “cut through” geographical levels and interact across these nested hierarchies (**lundvall1992?**; **carayannis2016?**).

Finally, this ecosystem framing directly addresses the concern raised by (**batheltstorper2023?**) that relatedness patterns can reflect outcomes (and size) more than the learning processes that produced them. Ecosystem approaches push attention toward the inputs and conditions of learning and experimentation—such as absorptive capacity, network connectivity, institutional support, and access to resources—which vary systematically across places and can be targeted by policy in ways that pure relatedness measures cannot (**qianacsstough2013?**; **stamwelter2020?**; **szerb2020?**). This is also consistent with the recent methodological move in the relatedness/complexity literature to evaluate measures by their predictive content and to complement proximity maps with richer, context-sensitive approaches, including machine-learning benchmarks and portfolio-based indicators of coherence and capability alignment—an approach we return to in later sections (**tacchella2023relatednessml?**; **albora2023productprogression?**; **straccamore2025fitnesscoherence?**).

Taken together, these arguments motivate the core move in this dissertation: a more contextualized assessment of diversification feasibility that keeps the strengths of the relatedness/network tradition (path dependence, structured opportunity sets) while explicitly incorporating (i) agency (firms and entrepreneurs as carriers of capabilities and novelty)

(Neffke et al. 2018), and (ii) institutional–system conditions approximated through entrepreneurial ecosystems and higher-level contexts that shape whether potential translates into sustained diversification (Hekkert et al. 2007; **stamwelter2020?**; **audretschbelitski2017?**). This provides a coherent bridge from an evolutionary resilience-adjacent framing to an empirical strategy that decomposes diversification into endogenous opportunity structure and exogenous enabling conditions—while remaining faithful to the EEG insight that regional change is path-dependent, multi-actor, and deeply place-contingent (P.-A. Balland, Rigby, and Boschma 2015; Boschma 2017; Neffke et al. 2018).

2.3 Relatedness as the baseline, context as the extension

The critiques raised by (**batheltstorper2023?**) and Whittle and Kogler (2020) find a rigorous empirical counterpart in recent methodological work. (**tacchella2023relatednessml?**) demonstrate systematically—across trade, patent, and scientific production data—that the co-occurrence-based relatedness measures that underpin the product space and related approaches perform worse in predicting new specialisations than trivial autocorrelation strategies: simply predicting that a region’s current basket will remain unchanged outperforms the Product Space in the COMTRADE dataset in multiple classification metrics. The reason is structural rather than incidental. The number of activities in any such dataset vastly exceeds the number of locations from which co-occurrences are counted, meaning that “the correlation structure that emerges is mostly random” (**tacchella2023relatednessml?**). Additionally, bipartite location-activity networks exhibit a strong nested structure, in which the relatedness signal is of second order

relative to the drive toward diversification—causing more diversified regions to score systematically higher on co-occurrence measures, independently of any genuine capability similarity (**tacchella2023relatednessml?**). This is precisely the “scale artifact” (**batheltstorper2023?**) identify: relatedness density conflates the size and diversity of a regional economy with the learning processes that generated it. The implication is that a purely dyadic proximity-to-entry account of diversification does not only miss mechanisms—it likely measures the wrong thing.

The methodological response to these limitations motivates a shift from two-body (pair-wise) to many-body (portfolio-level) measures of feasibility. (**tacchella2023relatednessml?**) show that moving from co-occurrence networks to decision-tree-based machine learning models—which condition predictions on the full activity basket rather than summing independent pairwise relatedness scores—improves forecasting precision by up to 384% over the Product Space in out-of-sample link prediction tasks. The core insight is that “describing countries’ development paths as the sum of binary Relatedness relationships is an oversimplification” (**tacchella2023relatednessml?**): what matters is not how close any single candidate technology is to any single existing one, but whether the portfolio as a whole provides the combinatorial conditions for a new specialisation to emerge. (**albora2023productprogression?**) operationalise a related approach via the Product Progression Probability (PPP), a random-forest-based method that uses the current basket as input to predict industrial upgrading, and show it substantially outperforms pairwise co-occurrence methods—while acknowledging that false positives remain common and statistical regularities in diversification are “simply too weak”

for confident individual-level prediction (C. Pinheiro 2025). Taken together, these methodological advances provide the empirical foundation for the present study’s use of a machine-learning-based potential measure trained on portfolio configurations rather than pairwise distances—treating feasibility as a property of the whole capability system, not of isolated technology-region pairs.

From an industrial policy perspective, the limitations of co-occurrence relatedness extend beyond measurement to theory. C. Pinheiro (2025) argues that recommending related diversification as a general strategy rests on three flawed premises: that related diversification is more common than unrelated diversification (empirically contested); that higher frequency implies higher success rates (not demonstrated); and that the observed pattern reflects a “natural law” of diversification rather than the accumulated effect of past policies that primarily incentivised related activities (Andreoni and Chang 2019). Most consequentially for the present study, C. Pinheiro (2025) draws on developmental state and innovation scholarship (amsden1989?; kimmelson2000?; wade2004?) to show that the most successful catching-up trajectories in history—South Korea, Ireland, Taiwan—involved precisely the “long jumps” into unrelated productions that REC-based frameworks classify as near-impossible. As Hidalgo et al. (2007) themselves acknowledged, “it is precisely these long jumps that generate subsequent structural transformation, convergence and growth.” This creates a direct tension for policy: relatedness-based recommendations may be appropriate for already-advanced regions with sufficient related capabilities to draw on, but they risk locking less developed regions into low-complexity trajectories. (pinheiro2025a?) confirm this empirically for European regions, showing that related

diversification has disproportionately benefited regions that were already advanced, generating a spatial inequality feedback loop. The implication is not that relatedness is irrelevant, but that its role as a constraint on feasibility is strongly conditioned by a region’s position in the capability distribution—and that measuring feasibility without accounting for this conditioning risks systematically underestimating the diversification potential of lagging regions while reinforcing the advantages of leading ones.

If relatedness captures the proximity dimension of the capability constraint, network structure captures the architectural dimension—specifically, the role that different positions in the technology space play in shaping which regions can access which diversification pathways. Tóth et al. (2022) operationalise this intuition using the technology space as a capability network, and Napolitano et al. (2018) show that constrained search processes through these networks generate systematic differences in regional development outcomes. The relevant insight here is that not all technologies are equivalent as diversification targets, even when controlling for relatedness: technologies that serve as bridges between otherwise disconnected knowledge domains—what the innovation literature terms General Purpose Technologies (**bresnahan1995?**; **helpman1998?**)—may generate qualitatively different diversification opportunities than technologies that are highly embedded in specific knowledge clusters. The former enable recombination across domains; the latter deepen existing specialisations. This distinction matters because, as C. Pinheiro (2025) argues, supply-side capability assessments alone cannot identify which diversification options are strategically valuable—and strategic value depends partly on where in the network architecture a technology sits. Mapping network positions (centrality, betweenness, eigenvector) onto

feasibility outcomes therefore provides a mechanism layer that relatedness density cannot supply: it specifies not only whether a technology is reachable from a region’s current portfolio, but what kind of recombination potential it carries and for which types of regions it is most accessible.

A further dimension absent from standard relatedness accounts is the coherence of the regional capability portfolio itself—the degree to which a region’s existing specialisations form a structurally consistent bundle rather than an eclectic collection. (teecerumelt1994?) introduced coherence in the context of firm diversification, defining it as the degree to which a firm’s activities draw on a common underlying set of capabilities. The concept translates directly to regions: a region whose technological portfolio is coherent—in the sense that its existing specialisations are embedded in the technology network in similar ways—has a capability base that is internally consistent and potentially more effective at supporting related recombination than a region whose portfolio spans disparate network positions, even if both have similar levels of relatedness density. This distinction matters because coherence captures a property of the whole portfolio that pairwise relatedness cannot: two regions can have the same average density vis-à-vis a candidate technology while differing substantially in whether their existing specialisations share the combinatorial infrastructure needed to support entry. This is not merely a refinement of the relatedness concept but a qualitatively different moderating condition—one that conditions how network positions translate into feasibility outcomes. As (tacchella2023relatednessml?) argue, “describing development paths as the sum of binary Relatedness relationships is an oversimplification”—coherence is precisely the

kind of higher-order portfolio property that many-body approaches can detect and that pairwise approaches necessarily miss.

BE CAREFUL REPETITION

The institutional dimension becomes central at this point not as background context but as a mechanism. Networks describe feasible recombinations; institutions condition whether those recombinations can be enacted and sustained. Neffke et al. (2018) make this mechanism explicit: access to capabilities depends on embeddedness in “economic, social, and trust relations with governments, institutions, and other firms,” implying two distinct channels—institutions shape local resource access through embeddedness, and they shape reliance on extra-local capabilities through connectivity to external networks. This is consistent with long-standing arguments that localised institutions, norms, and relational infrastructures condition learning, coordination, and innovation in places ([storper1995?](#); [cookemorgan1998?](#); [grabher1993?](#); [saxenian1994?](#)), and with the geography of innovation view that knowledge spillovers, interactive learning, and absorptive capacity are uneven across space ([cohenlevinthal1990?](#); [audretschfeldman1996?](#); [visserboschma2004?](#)). Critically, C. Pinheiro (2025) argues that REC-based frameworks are biased toward domestic supply, ignoring the demand-side and competitive dynamics that condition whether capability accumulation translates into successful diversification. ([leemalerba2017?](#)) formalise this concern by showing that “windows of opportunity” arise not only from technological paradigm shifts but also from shifts in de-

mand and from the changing competitive positions of rival economies—all of which are shaped by institutional environments. This motivates treating entrepreneurship and institutions as a coupled system rather than treating institutions as a single latent factor. The Global Entrepreneurship Index (GEI) operationalises this coupling through a systemic approach—combining individual and institutional variables—and applies a Penalty for Bottleneck logic that penalises underperforming pillars rather than allowing compensatory averaging (Acs, Autio, and Szerb 2014; **szerb2020?**). This is among the few ecosystem measures explicitly designed to capture both the individual and the systemic dimensions of the institutional enabling environment, making it a methodologically appropriate operationalisation of the higher-level context that the present study requires.

One dimension conspicuously absent from existing accounts of relatedness-constrained diversification is spatial interdependence. The standard capability network framing treats each region’s feasibility as a function of its own portfolio and the global technology network—implicitly assuming independence across regions. Yet evolutionary economic geography provides strong reasons to expect spatial clustering of diversification outcomes: knowledge spillovers are geographically bounded (**audretschfeldman1996?**), learning processes are proximity-dependent (**boschma2005?**), and the institutional and relational infrastructures that enable capability mobilisation are spatially embedded (**brownmason2017?**; **audretschbelitski2017?**). If neighbouring regions share similar capability bases, face similar institutional environments, or generate knowledge externalities that spill across borders, then their diversification trajectories will be correlated in

ways that a purely dyadic, aspatial framework cannot capture. P.-A. Balland, Rigby, and Boschma (2015) identify this as one of the underexplored dimensions of evolutionary resilience research: spatial structure shapes not only what regions can do but how their options co-evolve with those of neighbouring economies. This has a direct implication for empirical strategy: estimating diversification feasibility without accounting for spatial interdependence risks attributing to internal capability configurations what is partly the product of regional clustering and spatial spillovers. Incorporating spatial models—specifically spatial autoregressive and spatial error specifications—therefore constitutes not merely a robustness check but a substantive test of whether the feasibility landscape is spatially structured and, if so, of how much of regional diversification potential is shaped by neighbourhood effects versus internal portfolio characteristics.

Taken together, these arguments motivate the core empirical move of this dissertation: a more contextualised assessment of diversification feasibility that keeps the strengths of the relatedness and network tradition—path dependence, structured opportunity sets, capability constraints—while explicitly incorporating the dimensions that this tradition has neglected. The first of these is the measurement dimension: moving from pairwise co-occurrence proximity to portfolio-level, machine-learning-based feasibility assessment that captures higher-order capability configurations (**tacchella2023relatednessml?; albora2023productprogression?**). The second is the network architecture dimension: decomposing technology positions into bridging (betweenness) and authority (eigenvector) roles that carry qualitatively different implications for which regions can access which diversification pathways. The third is the coherence dimension: treating portfolio-level

structural consistency as a moderating condition that shapes how network positions translate into feasibility outcomes (teecerumelt1994?). The fourth is the institutional-ecosystem dimension: approximating the enabling conditions for capability mobilisation through a systemic, bottleneck-sensitive measure of national entrepreneurial ecosystems (Acs, Autio, and Szerb 2014; szerb2020?)—addressing C. Pinheiro (2025)’s concern that supply-side capability maps miss the demand, competition, and institutional conditions that determine whether feasibility becomes realised diversification. The fifth is the spatial dimension: testing whether the feasibility landscape is spatially structured, which P.-A. Balland, Rigby, and Boschma (2015) identify as a neglected but theoretically motivated dimension of evolutionary regional change. This five-part decomposition is not an arbitrary accumulation of controls—it is a theoretically grounded architecture that maps directly onto the gaps identified across the evolutionary, complexity, and industrial policy literatures, and that together allows a more complete account of what shapes regional technological diversification than any single-dimension approach can provide.

3 The interconnectedness between regions and technologies

3.1 Methodological Motivation

Traditional relatedness frameworks model diversification through symmetric co-occurrence matrices and linear aggregation of relatedness density (Hausmann and Hidalgo 2011). However, as established in our problem statement, these constructs suffer from three structural limitations: (1) symmetry assumptions that obscure directional dependencies between technologies, (2) linear aggregation that loses granular information about technology-specific requirements, and (3) noise when technologies outnumber regions.

We address these limitations by replacing traditional measures with a machine learning training strategy based on the Random Forest algorithm (RF). Specifically, we use RF to generate: (1) **Feature Importance Technology Space (FITS)** following Fessina et al. (2024), analogous to the mainstream Technology Space. FITS captures directional, hierarchical technology relationships, replacing symmetric relatedness measures; and (2)

predicted probabilities (technological potential) following Albora et al. (2023), analogous to relatedness density. The technological potential estimates region-specific feasibility of technology adoption given the region’s own technological portfolio, replacing linear relatedness density. This approach enables the contextualization of diversification strategies by accounting simultaneously for technology-specific characteristics, regional knowledge infrastructure, and the broader non-linear interconnectedness between regions and technologies that traditional measures cannot adequately capture.

Before detailing our methodology and how we leverage the contributions in Fessina et al. (2024) and Albora et al. (2023), we will briefly situate this approach within the REC literature. Essentially, REC formalizes the empirical observation that shared input requirements (knowledge, resources, capabilities) determine diversification feasibility (Hidalgo et al. 2018; Hidalgo 2021). While this framework have proven valuable for policy (Zaccaria et al. 2018; E and A 2021), deriving granular, context-specific implications remains challenging (Hidalgo 2023; Li and Neffke 2024). Recent work on unrelated diversification (Flávio L. Pinheiro et al. 2022; Boschma et al. 2023), geographic inequalities (Flavio L. Pinheiro et al. 2025; Hartmann et al. 2017), and emerging technologies (Lee et al. 2018; Fessina et al. 2024) highlights the need for methodologies that capture contextual nuances beyond path dependency identification which we will elaborate on in subsequent sections.

Our approach is a response to the nuances expressed in the literature. But in order to highlight adequately our work we must first elaborate on the mainstream approach conceptually, methodologically and its empirical implications which we briefly outlined in the introduction. From there we expand on the mainstream approach and construct ours

conceptually, mathematically and showcase our training strategy. To this end, we also provide a brief example on how the mainstream approach and the machine learning approach to modeling the technology space compare. For the sake of clarity and consistency we will establish our notation system and the main definition in this paragraph and use them in the subsequent sections and chapter in this dissertation unless we specify otherwise.

We consider the following sets:

- $\mathcal{T} = \{t_i\}_{i=1}^{N_T}$: set of technologies
- $\mathcal{R} = \{r_i\}_{i=1}^{N_R}$: set of regions
- $\mathcal{Y} = \{y_i\}_{i=1}^{N_Y}$: set of years
- $\mathcal{C} = \{c_i\}_{i=1}^{N_C}$: set of technology categories
- $\mathcal{K} = \{\kappa_i\}_{i=1}^{N_K}$: set of countries

These sets are mirrored by the European Patent Office data (1978-2021) classified at the 4-digit IPC level, yielding 641 distinct technologies across 345 NUTS2 regions in 34 European countries. IPC classifications provide hierarchical structure: section (letter, we also refer to this as categories), class (two digits), subclass (letter). For example, F16H encompasses Section F (Mechanical engineering), Class 16 (engineering elements for mechanical power transmission), and Subclass H (gearing systems). We supplement patent data with Eurostat regional socio-economic indicators detailed in subsequent sections.

We quantify regional specialization using the Revealed Comparative Advantage (Balassa 1965). Although originally designed for trade data, this metric has been widely adopted

in innovation geography literature. Following the nomenclature in P.-A. Balland and Rigby (2017), we refer to it as Revealed Technological Advantage (RTA) in our patent data context. The RTA measures relative specialization, enabling simultaneous capture of expertise depth and portfolio diversity. Despite critiques regarding patent-based applications (P. Balland and Boschma 2019; Diodato et al. 2023), RTA aligns with our objective of capturing meaningful technology relationships through machine learning rather than raw co-occurrence. The RTA for region r in technology t during year y is defined as:

$$\text{RTA}_{r,t,y} = \frac{\frac{X_{r,t,y}}{\sum_{t'} X_{r,t',y}}}{\frac{\sum_{r'} X_{r',t,y}}{\sum_{r',t'} X_{r',t',y}}} = \frac{X_{r,t,y} \sum_{r',t'} X_{r',t',y}}{(\sum_{t'} X_{r,t',y}) (\sum_{r'} X_{r',t,y})}$$

Where, $X_{r,t,y}$: patent count for region r in technology t during year y

For each year y , we construct the RTA matrix $\mathbf{R}^{(y)}$ with entries $\text{RTA}_{r,t,y}$:

$$\mathbf{R}^{(y)} = [\text{RTA}_{r,t,y}]_{r=1,\dots,N_R}^{t=1,\dots,N_T}$$

These yearly matrices form the foundation for all subsequent modeling. We then binarize specialization for classification tasks:

$$z_{r,t,y} = \begin{cases} 1 & \text{if } \text{RTA}_{r,t,y} \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

where $z_{r,t,y} = 1$ indicates region r has comparative advantage (specialization) in technol-

ogy t at year y .

To further elaborate, let's consider $c_{t,t'}$, the count of patents containing both technologies t and t' ($t \neq t'$) at a given time period, such that $t, t' \in \mathcal{T}$, and $c_t, c_{t'}$ denote the total patent counts for each class individually.

In the following we will begin by outlining how the REC literature approach relatedness, how it's calculated and discuss in detail different technical contributions and their limitations. We will then present our approach and detail our methodological and conceptual contribution in response to these limitations.

3.2 The mainstream approach

3.2.1 Relatedness Through Co-occurrence

The REC literature operationalises relatedness through the observed frequency of co-occurrences. The intuition is appealingly simple: technologies that frequently appear together—whether in the same patent, the same firm, or the same region—likely share underlying knowledge requirements. If mastering one technology facilitates developing another, we should observe them co-located more often than chance would predict. This reasoning has proven influential, providing the empirical foundation for technology space visualisations and diversification recommendations across academic and policy settings.

Yet the inference from co-occurrence to capability-based relatedness rests on assumptions that warrant scrutiny. Co-occurrence is a frequency measure; it tells us how often two

technologies appear together, not why they do so or whether that co-presence reflects shared inputs, complementary applications, or statistical artefact. The distinction matters: a measure intended to guide capability-building strategies must capture genuine knowledge relationships rather than patterns that emerge from data structure alone.

Let $c_{t,t'}$ denote the count of patents containing both technologies t and t' (where $t \neq t'$) within a given time period, and let c_t and $c_{t'}$ denote the total patent counts for each technology individually. The simplest relatedness measure would be $c_{t,t'}$ itself, but raw co-occurrence counts are problematic: frequently patented technologies will naturally co-occur more often simply due to their prevalence. Various normalisation schemes address this issue.

Association Strength compares observed co-occurrence to expected co-occurrence under independence:

$$\phi_{t,t'}^{\text{assoc}} = \frac{c_{t,t'} \cdot T}{c_t \cdot c_{t'}}$$

where $T = \sum_{t \in \mathcal{T}} c_t$ is the total patent count. Values exceeding one indicate that the technologies appear together more frequently than would be expected if their occurrences were independent—they are positively associated. This measure properly corrects for size effects: a rare technology and a common technology can still exhibit strong association if they co-occur more than their marginal frequencies would predict.

The Probability Index refines this probabilistic approach by accounting for the fact that a patent cannot co-occur with itself:

$$\phi_{t,t'}^{\text{prob}} = \frac{c_{t,t'}}{\frac{1}{2} \left[\frac{c_t}{T} \cdot \frac{c_{t'}}{T-c_t} + \frac{c_{t'}}{T} \cdot \frac{c_t}{T-c_{t'}} \right] \cdot T}$$

This correction, while subtle, provides more accurate estimates for patent co-occurrence analysis and is implemented in standard software packages used in the field.

Cosine Similarity offers a set-theoretic alternative:

$$\phi_{t,t'}^{\text{cosine}} = \frac{c_{t,t'}}{\sqrt{c_t \cdot c_{t'}}}$$

This measures relative overlap between technology profiles. While intuitive, it systematically favours frequent technologies over rare ones and does not properly correct for size effects, making comparisons across technologies with different prevalences problematic.

Minimum Conditional Probability takes a different approach by examining regional co-location rather than patent-level co-occurrence:

$$\phi_{t,t'}^{\text{min}} = \min \{P(\text{RTA}_{r,t}|\text{RTA}_{r,t'}), P(\text{RTA}_{r,t'}|\text{RTA}_{r,t})\}$$

This computes the minimum of the conditional probabilities: given that a region specialises in one technology, how likely is it to specialise in the other? The minimum operator ensures symmetry while avoiding inflation from imbalanced conditional relationships.

A clarification is warranted here regarding two distinct but often conflated concepts. Co-occurrence typically refers to technologies appearing together within the same patent or

firm, while co-location refers to technologies appearing together within the same region. Both capture related but different phenomena: co-occurrence within patents reveals direct technical complementarities, while co-location within regions reveals shared capability requirements that may operate at greater conceptual distance. The literature sometimes uses these terms interchangeably, but the distinction matters for interpreting what relatedness measures actually capture.

3.2.2 Structural Limitations of Co-occurrence-Based Relatedness

Despite the variety of normalisation approaches, co-occurrence-based relatedness measures share structural limitations that affect their reliability and interpretation.

Dimensional mismatch produces noisy estimates. The reliability of frequency-based measures depends on having sufficient observations to estimate co-occurrence patterns accurately. In typical applications, the number of technologies substantially exceeds the number of locations—estimating a matrix of several thousand technology pairs from fewer than two hundred country observations, for instance, produces correlation structures that are largely random (Tacchella et al. 2023). The number of observations required for reliable estimation of such matrices easily reaches tens of thousands (Bun, Bouchaud, and Potters 2017), far exceeding available data. This problem is not resolved by moving to subnational levels, where harmonised data is often unavailable and the fundamental dimensional mismatch persists.

Nested structure dominates the relatedness signal. Location-activity networks typically exhibit strong nested structure: diversified regions produce almost everything while

less diversified regions produce only ubiquitous activities (Mariani et al. 2019). This triangular pattern means that when counting co-occurrences, one primarily detects the diversification hierarchy rather than genuine capability-based relationships. The relatedness signal becomes second-order relative to the structural pattern that generates nestedness (Tacchella et al. 2023). This explains why even randomised relatedness matrices produce predictions comparable to co-occurrence-based measures—they still capture the fact that diversified regions are likely to add activities regardless of which specific activities are involved (Bustos et al. 2012).

Spurious associations are rarely filtered. Studies employing co-occurrence measures often fail to test the statistical significance of the relationships they identify (Saracco et al. 2015, 2017). Without appropriate null models, spurious co-occurrences due to sampling variability are mistaken for genuine relatedness. When such tests are applied to trade data, the results are sobering: the null hypothesis of random diversification—no path dependence—is rejected for only approximately half of countries examined, with most high-income and large economies exhibiting diversification patterns that defy path dependence entirely (Coniglio et al. 2021). Whether these findings extend to patent-based technology data remains an open question, but they raise concerns about the generalisability of path-dependent diversification as a universal empirical regularity.

Symmetry obscures directional dependencies. All normalisation schemes above produce symmetric measures: $\phi_{t,t'} = \phi_{t',t}$. This symmetry assumes that the relationship between any two technologies operates bidirectionally with equal strength. Yet technological development often exhibits clear hierarchies (Boschma 2017). Expertise in semicon-

ductor fabrication may strongly predict future development of integrated circuit design, but the reverse need not hold—circuit designers may not subsequently develop fabrication capabilities. The capabilities required to transition from technology i to technology j may differ vastly from those required for the reverse transition. Symmetric measures cannot distinguish prerequisite relationships from derived relationships, enabling technologies from their applications, or identify the stepping-stone pathways that matter for strategic planning.

3.2.3 Relatedness Density Through Linear Aggregation

Relatedness density (Hidalgo et al. 2007) bridges the gap between pairwise technology relationships and region-specific diversification potential. Where co-occurrence measures describe technology-to-technology proximity, relatedness density aggregates these proximities relative to a region’s existing portfolio, producing a single value meant to capture how feasible a new technology is for a given region. The measure has become central to diversification analysis and policy guidance, serving as the primary predictor in entry models and the basis for strategic recommendations.

The aggregation, however, imposes strong assumptions about how capabilities combine. By summing pairwise relationships, relatedness density treats the regional portfolio as a linear accumulation of independent components. Whether this adequately represents how knowledge and capabilities actually interact in enabling new activities is an empirical question that the measure’s structure forecloses.

For region r and technology t , relatedness density aggregates the relatedness between technology t and all technologies in which the region currently specialises:

$$\text{relatedness density : } \Phi_{r,t} = \frac{\sum_{t' \in \mathcal{T}_r, t' \neq t} \phi_{t,t'}}{\sum_{t' \neq t} \phi_{t,t'}} \times 100$$

where $\mathcal{T}_r$ denotes the set of technologies in which region r specialises (those with $\text{RTA}_{r,t} \geq 1$).

The interpretation is intuitive: the numerator sums relatedness values to technologies the region already has, while the denominator normalises by total relatedness to all technologies. High relatedness density indicates that technology t is “close” to the region’s existing capabilities—many of its related technologies are already present in the regional portfolio. Low relatedness density suggests that t requires capabilities the region currently lacks.

This measure has proven remarkably predictive: across numerous studies using different data, time periods, and geographic contexts, higher relatedness density consistently associates with higher probability of future entry. Regions are more likely to develop specialisations in technologies that relate to what they already do well.

3.2.4 Structural Limitations of Linear Aggregation

The predictive success of relatedness density is not in question. The limitations concern what it cannot reveal and how its structure constrains interpretation.

Linear aggregation obscures combinatorial information. Relatedness density sums

pairwise relationships without regard to their interactions. This implicitly assumes that having two related technologies is simply twice as beneficial as having one—capabilities combine additively. Yet capabilities may interact in ways that linear summation cannot capture. Certain combinations may be more than the sum of their parts, creating complementarities where the joint presence of technologies A and B enables technology C in ways that neither alone provides. Other combinations may be redundant, contributing less than their individual relatedness values would suggest. By reducing many-body relationships to independent two-body terms, linear aggregation discards precisely the combinatorial information that may distinguish feasible diversification paths from infeasible ones (Tacchella et al. 2023). Empirically, prediction models that capture many-product interactions substantially outperform density-based predictions, suggesting that linear aggregation loses significant information (Albora et al. 2023).

Noise compounds through aggregation. The dimensional mismatch and spurious associations affecting pairwise co-occurrence estimates do not disappear when aggregated into relatedness density—they compound. Summing noisy pairwise values produces a noisy aggregate. The normalisation by total relatedness provides some correction but cannot recover signal that was lost at the pairwise level. For regions with portfolios concentrated in rare technologies, where pairwise estimates are least reliable, relatedness density inherits and potentially amplifies these estimation problems.

Frequency of entry conflates observation with feasibility. The empirical regularity that higher relatedness density associates with higher entry probability is often interpreted as evidence that related diversification is more feasible and therefore should be priori-

tised. But this inference conflates observed frequency with success rate as emphasised in C. Pinheiro (2025). We observe which entries occurred, not which were attempted; the denominator—attempts—remains unobserved. Related diversification may appear more common because it is attempted more frequently, perhaps precisely because existing frameworks recommend it, rather than because it succeeds more often when attempted. Coniglio et al. (2021) addresses precisely this issue and shows that related diversification attempts succeeded 80% in Germany in contrast to 40% in the United States and France. Moreover, the observed product and technology spaces may themselves reflect past policies promoting related diversification (Andreoni and Chang 2019), creating circularity when used to justify future policy.

3.2.5 Implications for Contextualisation

The limitations identified above are not independent—they compound. Noisy co-occurrence estimates feed into symmetric matrices that are then linearly aggregated, producing context-free measures whose predictive success may reflect structural properties of the data (nestedness, diversification hierarchies) as much as genuine capability-based relationships. The resulting framework provides a supply-side instrument: it identifies what regions might produce given their existing portfolios, while remaining silent on demand dynamics, competitive pressures, and the institutional conditions that shape whether capability-based proximity translates into actual entry (C. Pinheiro 2025; Andreoni and Chang 2019).

This supply-side orientation has consequences. When applied to policy, relatedness-based

recommendations may systematically favour already-diversified regions whose dense portfolios generate high relatedness density across many targets, while offering limited guidance to less-diversified regions whose sparse portfolios produce low density values regardless of strategic potential. The concern that such recommendations reinforce rather than ameliorate path dependency—locking peripheral regions into constrained trajectories while core regions accumulate further advantages—has been explicitly raised in recent literature examining regional inequality and the distributional consequences of relatedness-based strategies (Flavio L. Pinheiro et al. 2025; Mealy and Coyle 2022; Hidalgo 2023).

The issue is not that relatedness is empirically wrong—the predictive regularity is robust—but that the measures as currently constructed cannot inform us about the conditions under which relatedness matters more or less, about which technologies serve as stepping stones toward others, or about how regional knowledge infrastructure, national ecosystems, and spatial factors moderate diversification possibilities. Addressing these limitations requires reconstructing the core methodological constructs in ways that relax the problematic assumptions while preserving the fundamental insight that prior capabilities shape future possibilities. This reconstruction motivates the machine learning approach developed in the following sections.

3.3 An Alternative Approach: Machine Learning for Contextualised Relatedness

The limitations outlined above—dimensional mismatch producing noisy correlations, symmetric measures obscuring directional dependencies, and linear aggregation discarding interaction effects—are not merely technical inconveniences. They represent fundamental obstacles to using relatedness for prediction and policy. If co-occurrence methods perform no better than trivial baselines, as demonstrated in Tacchella et al. (2023), then their assessment of diversification feasibility offers insufficient guidance for strategic decisions. This motivates a shift from descriptive correlation to predictive modelling.

3.3.1 Machine Learning as Prediction Framework

The core reframing is methodological: rather than inferring relatedness from co-occurrence patterns and hoping this correlates with future outcomes, we directly model diversification as a prediction problem. This follows recent work positioning relatedness estimation within supervised learning frameworks (Tacchella et al. 2023; Albora et al. 2023; Fessina et al. 2024). The shift offers three advantages.

First, it provides a falsifiable evaluation standard. Co-occurrence measures proliferate without systematic comparison—proximity, association strength, minimum conditional probability, and numerous variants coexist with no principled basis for selection. Out-of-sample prediction performance offers an objective criterion: methods that better forecast actual diversification outcomes are, by definition, better capturing whatever underlying

relatedness structure matters for real transitions (Albora et al. 2023). This addresses the circularity where relatedness is inferred from outcomes and then used to explain those same outcomes.

Second, prediction-based approaches can capture the many-body correlations that density-based measures discard. As Tacchella et al. (2023) emphasise, describing diversification paths as sums of binary relatedness relationships is an oversimplification. A region’s probability of developing technology i depends not just on pairwise similarities but on the full configuration of its existing portfolio—which combinations are present, which are absent, and how these interact. Tree-based algorithms learn precisely these complex conditional patterns.

Third, the framework naturally accommodates the asymmetric, directed relationships that symmetric co-occurrence measures cannot represent. When we train separate models for each target technology, we obtain directional importance scores: technology t may strongly predict entry into technology i while i weakly predicts entry into t . This asymmetry reveals hierarchical structure—prerequisites, enabling technologies, and developmental sequences—that bidirectional similarity metrics obscure.

We apply this framework through two complementary constructs:

Technological Potential estimates $p_{r,t,y}$, the probability that region r develops specialisation in technology t by year y , given its existing portfolio. This replaces relatedness density. Rather than linearly summing symmetric co-occurrence frequencies, we predict region-specific feasibility using the full structure of past portfolios. The measure captures non-linear interactions and regional heterogeneity that additive approaches miss.

Feature Importance Technology Space (FITS) extracts directional dependencies $I_{t \rightarrow t'}$ between technologies from the trained models. This replaces symmetric technology networks. Instead of co-occurrence frequencies, we identify which technologies predict others’ future adoption—and crucially, which do not predict in the reverse direction. The resulting asymmetric network reveals hierarchical technological structure invisible to symmetric measures.

3.3.2 Why Random Forest

Among supervised learning approaches, tree-based ensemble methods—particularly Random Forest—offer properties well-suited to relatedness estimation. We briefly outline these before presenting the technical details.

High-dimensional sparse data. Regional technology portfolios are high-dimensional (hundreds of technology classes) and sparse (most regions specialise in few technologies). Tree-based methods handle this naturally through recursive partitioning: each split considers only a subset of features, and the hierarchical structure means irrelevant technologies simply do not appear in decision paths. Linear methods struggle with the collinearity and sparsity typical of specialisation data; neural networks require larger samples than regional patent data typically provide.

Non-linear interactions. The recursive splitting structure captures interactions without explicit specification. If technology t_1 matters for predicting t_3 only when t_2 is also present, this emerges naturally as a conditional split pattern. Density-based measures, by

construction, cannot represent such conditionality—they assume each related technology contributes independently.

Interpretable feature importance. Unlike black-box models, Random Forests produce feature importance scores that aggregate each predictor’s contribution across all trees. These scores become the basis for FITS construction, translating prediction performance into an interpretable technology network. This maintains the explanatory value of relatedness measures while grounding them in predictive validity.

Ensemble variance reduction. Bootstrap aggregation across many trees reduces the variance that would plague individual decision trees, particularly given the moderate sample sizes (regions $\times$ years) available in patent data. The ensemble structure provides stability without sacrificing the flexibility needed to capture complex capability interactions.

The choice of Random Forest over alternatives thus reflects both practical constraints (sample size, interpretability requirements) and the specific structure of relatedness estimation (sparse, high-dimensional, interaction-rich). We now present the algorithm formally before describing our application.

Split quality is assessed via Gini impurity:

$$G(t) = 1 - \sum_{k=0}^1 p_k^2(t) = 2p_0(t)p_1(t)$$

where $p_k(t) = n_k(t)/n(t)$ represents the proportion of class k observations at node t . Gini impurity quantifies node heterogeneity: $G = 0$ indicates perfect purity (homogeneous class), while $G = 0.5$ indicates maximum impurity (equal class distribution). The optimal split maximises weighted impurity reduction:

$$\Delta G(j, \tau) = G(t) - \left[\frac{n(t_L)}{n(t)} G(t_L) + \frac{n(t_R)}{n(t)} G(t_R) \right]$$

Weighting by relative node size prevents trivial splits that isolate single observations into pure but uninformative leaves.

Recursive splitting continues until predefined stopping criteria: node purity ($G = 0$), minimum node size threshold, or maximum tree depth. Terminal nodes are assigned the majority class of their constituent observations.

For prediction, observation $\mathbf{x}$ traverses all B trees. Final classification aggregates individual tree predictions via majority voting:

$$\hat{y}(\mathbf{x}) = \text{mode}\{\hat{y}_1(\mathbf{x}), \dots, \hat{y}_B(\mathbf{x})\}$$

Class probabilities are estimated as the proportion of trees predicting each class:

$$\hat{P}(y = 1|\mathbf{x}) = B^{-1} \sum_{b=1}^B \mathbb{I}[\hat{y}_b(\mathbf{x}) = 1]$$

This probability represents empirical vote share across trees. Values near 1 indicate strong consensus for class 1 (high confidence), while values near 0.5 reflect uncertainty. Unlike parametric models, these are data-driven vote proportions rather than model-based probability estimates.

The algorithm's effectiveness stems from variance reduction through decorrelated predictions. Bootstrap sampling and random feature selection reduce inter-tree correlation ρ , yielding ensemble variance:

$$\text{Var}(\bar{y}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

As B increases and ρ decreases, ensemble variance diminishes while maintaining the low bias of flexible tree models.

Feature importance quantifies each predictor's contribution by aggregating Gini impurity reductions:

$$I(j) = \frac{1}{B} \sum_{b=1}^B \sum_{t \in T_b: v(t)=j} \Delta G(t)$$

where the sum runs over all nodes using feature j for splitting. Higher values indicate features consistently creating purer partitions. Feature importance measures predictive

association rather than causal effect, and exhibits bias toward high-cardinality features and correlated predictor sets. It ranks predictive relevance but requires caution in causal interpretation.

3.3.4 Training Procedure

We follow the methodology of Fessina et al. (2024) and Albora et al. (2023), originally developed for trade data. The key innovation lies in training separate models for each technology rather than constructing a single global model. For every technology $i \in \mathcal{T}$, we train a binary classification model where:

- **Outcome:** $z_{r,i,y}$ indicates whether region r specialises in technology i at year y ($\text{RTA} \geq 1$)
- **Features:** $\{\text{RTA}_{r,t,y-\delta} : t \in \mathcal{T}, t \neq i\}$ comprises RTA values for all other technologies δ years prior

This technology-specific approach enables each model to learn unique dependency patterns. The lag $\delta = 4$ years balances predictive horizon with data availability, following conventions in the literature on capability accumulation (Andreoni and Chang 2019).

We predict specialisation for target years $y_t \in \{2008, \dots, 2018\}$ using data from 1978 onward. For each target year y_t and technology i :

Training set:

$$X_{\text{train}} = \{\text{RTA}_{r,t,y} \mid y \in [1978, y_t - 2\delta], t \neq i\}$$

$$Y_{\text{train}} = \{z_{r,i,y} \mid y \in [1978 + \delta, y_t - \delta]\}$$

Test set:

$$X_{\text{test}} = \{\text{RTA}_{r,t,y_t-\delta} \mid t \neq i\}$$

$$Y_{\text{test}} = \{z_{r,i,y_t}\}$$

This temporal structure ensures strict separation: models predict future specialisation using only information available at the prediction date. The expanding training window incorporates historical diversification patterns while the fixed lag prevents information leakage.

The procedure yields 7,051 models (641 technologies $\times$ 11 years). Given computational constraints, we performed cross-validation on a stratified sample of four technologies (G06G, B67B, D02J, C08J) spanning different IPC sections, applying the modal optimal parameters across all models: `mtry = 139`, `trees = 100`, `min_n = 38`. Training was conducted in R using the Ranger package (Wright and Ziegler 2017), orchestrated via the targets pipeline (Landau 2021).

3.4 Technological Potential

Each model produces probabilities $p_{r,i,y} = P(z_{r,i,y} = 1 \mid \text{RTA}_{r,\cdot,y-\delta})$ representing the likelihood that region r develops specialisation in technology i by year y , conditional on

its prior portfolio. These probabilities constitute **Technological Potential**—a forward-looking, region-specific measure of diversification feasibility.

The shift from correlation to prediction reframes what the measure captures. Density asks: “How similar is technology i to what region r already has?” Potential asks: “Given everything we know about how regions diversify, how likely is region r to develop technology i ?” The latter question—the one relevant for policy—requires a predictive approach.

High potential ($p_{r,t,y} \approx 1$) indicates a region’s existing portfolio strongly predicts future specialisation—the region likely possesses necessary complementary capabilities. Low potential ($p_{r,t,y} \approx 0$) suggests capability gaps that make diversification unlikely under current conditions. Intermediate values reflect genuine uncertainty, often indicating that diversification is possible but contingent on factors beyond current portfolio composition.

Crucially, potential varies across regions for the same technology. Two regions with similar aggregate relatedness density may have quite different potential scores because their specific portfolio configurations interact differently with the target technology’s requirements. This variation enables analysis of how regional context—knowledge infrastructure, institutional environment, spatial position—moderates diversification feasibility.

3.5 Feature Importance Technology Space (FITS)

Traditional technology networks rely on patent citations or co-occurrence patterns, each with limitations. Citations suffer from examiner additions that may not reflect actual

knowledge flows, aggregation from patent-level to technology-level that obscures directionality, and backward-looking orientation that captures historical influence rather than predictive relationships (Fessina et al. 2024). Co-occurrence networks inherit the problems outlined in our critique: symmetry, noise, and linear assumptions.

FITS addresses these by constructing an asymmetric, predictive network from the feature importance scores of our Technological Potential models. Rather than inferring relationships from co-occurrence, FITS extracts directional dependencies revealed by which technologies predict others' future adoption.

Recall that for each technology i , we trained a Random Forest model predicting $z_{r,i,y}$ using $\{\text{RTA}_{r,t,y-\delta} : t \neq i\}$ as features. The feature importance $I_i(t)$ quantifies how much technology t contributes to predicting future specialisation in technology i across all regions and time periods.

We formalise FITS as a directed, weighted network $G = (V, E, W)$ where:

- **Nodes** $V = \mathcal{T}$: the set of 641 technologies
- **Edges** E : directed connections $(t \rightarrow i)$ for all $t, i \in \mathcal{T}, t \neq i$
- **Weights** $W_{t \rightarrow i} = I_i(t)$: feature importance of technology t in the model predicting technology i

We normalise weights within each target technology:

$$W_{t \rightarrow i} = \frac{I_i(t)}{\sum_{t' \neq i} I_i(t')}$$

ensuring incoming edge weights sum to 1 for each technology. This normalisation facilitates comparison across technologies with different overall predictability.

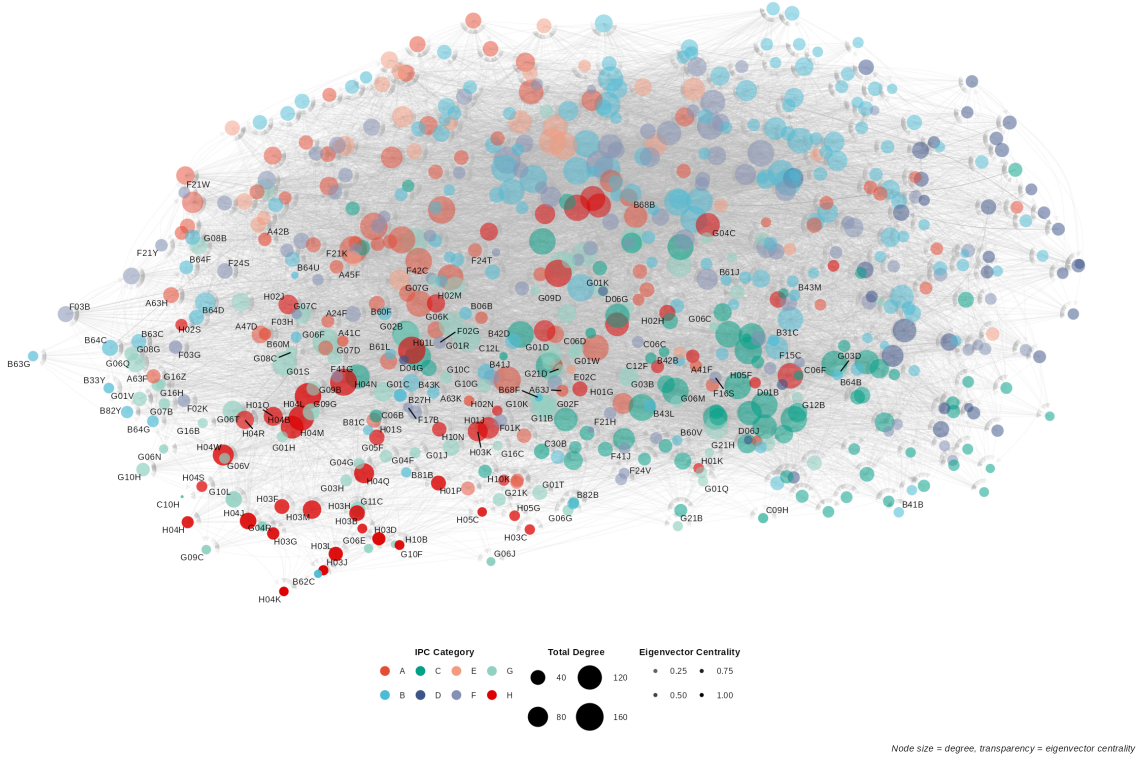


Figure 3.1: Technology Network (based on 2018 patent data)

The defining property of FITS is asymmetry: $W_{t \rightarrow i} \neq W_{i \rightarrow t}$ in general. This asymmetry captures hierarchical technological dependencies invisible to symmetric measures:

- $W_{t \rightarrow i} \gg W_{i \rightarrow t}$: Technology t is a prerequisite or enabler for i . Expertise in t predicts future development of i , but not vice versa. This pattern identifies foundational technologies and developmental sequences.
- $W_{t \rightarrow i} \approx W_{i \rightarrow t}$: Technologies are complementary peers with mutual predictive relationships, suggesting shared capability requirements or co-evolution.

- $W_{t \rightarrow i} \ll W_{i \rightarrow t}$: Technology i enables t , reversing the hierarchical relationship.

This directional structure reveals technological trajectories: regions can identify which current capabilities open paths toward desired future technologies, and which apparent similarities mask asymmetric dependencies.

The contrast between co-occurrence relatedness and FITS is sharpest when examining specific technologies. Consider H01L (semiconductors) and D06Q (textile decoration).

For semiconductors, co-occurrence relatedness identifies technologies that frequently appear together in regional portfolios. Some connections are sensible: semiconductor memories (H10B), packaging technologies (B68B). But the measure also surfaces implausible links—textile decoration (D06Q), woodworking (B27H), fusion reactors (G21B)—that appear not because of genuine technological affinity but because they happen to co-occur in diverse portfolios. This is the noise problem: frequency-based aggregation treats all co-presence equally, regardless of whether it reflects shared capabilities or statistical coincidence.

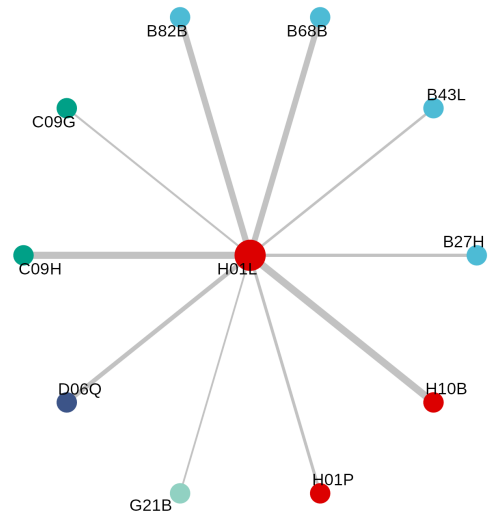
FITS reveals a different structure. Rather than asking which technologies appear alongside semiconductors, it asks which technologies predict future semiconductor development—and which technologies semiconductors predict. The asymmetry is striking: strong edges flow FROM application domains (digital computing G06F, telecommunications H04L, control systems G05F) TO semiconductors, indicating these fields depend on semiconductor capabilities. Weaker reverse edges show that semiconductor expertise does not particularly predict entry into computing or telecommunications. H01L emerges as an enabling technology whose importance lies in what it makes possible rather than what leads to it.

The contrast sharpens further with D06Q (textile decoration), a sparse technology where co-occurrence produces near-total noise: elevators (B64B), timing devices (G04D), chemical production (C13B) appear simply because D06Q is rare and co-occurs randomly with whatever else rare-technology regions happen to have. FITS filters this effectively, identifying genuine textile domain relationships—sewing (D05B), textile treatment (D06M), wall coverings (D06N)—that share actual functional connections.

This example illustrates how FITS captures the functional architecture of innovation networks that co-occurrence frequencies obscure. The method’s ability to distinguish signal from statistical artifact, combined with its directional structure revealing technological hierarchies, provides the foundation for contextualised analysis of regional diversification patterns.

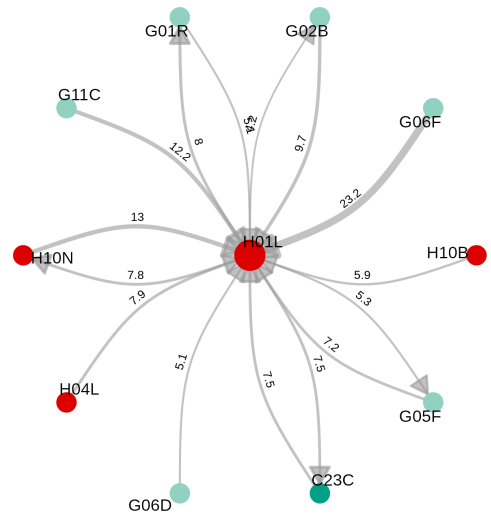
Relatedness Network

Co-occurrence based relationships to H01L



FITS Network

Bidirectional predictive relationships with H01L



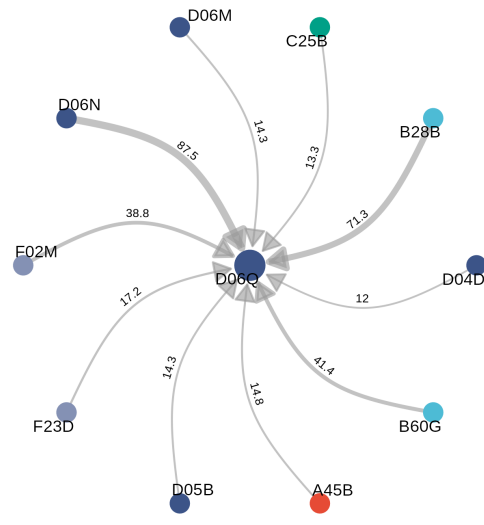
Relatedness Network

Co-occurrence based relationships to D06Q



FITS Network

Bidirectional predictive relationships with D06Q



Technological Potential provides region-specific feasibility estimates grounded in predictive performance rather than descriptive correlation. FITS reveals the directional, hierarchical structure of technological dependencies that symmetric measures cannot represent. Together, these constructs enable the contextualised analysis our research questions require: examining how regional knowledge infrastructure, national innovation systems, and spatial factors moderate the relationship between technological characteristics and diversification outcomes.

3.6 Coherence: Bridging Technology Networks and Regional Portfolios

Till now we established few points. First, FITS identifies directed technology-to-technology dependencies. And second, Regional Technological Potential quantifies region-specific feasibility of future specialisation. To bridge these two layers of information we introduce here the Coherence measure. Coherence here links these two levels by measuring whether a focal technology’s network profile matches the aggregate network profile of the region’s existing portfolio.

This logic follows the core idea in the coherent diversification tradition as established in (teece1994understanding?) and later in (nesta2005coherence?). The argument here is that performance is shaped not only by “how much” an actor does (diversification/variety), but also by how the activities in its portfolio hang together. Regional coherence in this sense is a structured property of the knowledge base, or as we referred to in the introduction as knowledge infrastructure. This logic aligns with recombinant views of innovation in which knowledge advances through combinations among related elements rather than as a homogeneous “stock” input (quatraro2010?). At the firm level, the same conceptual point is made sharply: two firms with the same number of technological fields can differ substantially depending on whether those fields form a coherent block or scattered knowledge infrastructure, coherence captures that internal structure (pugliese2019?).

3.6.1 Why we introduce coherence here (and why at IPC-section scale)

A recurring open issue in the literature is that the meaning and effect of coherence depend on scale and context. Recent studies shows that coherence can correlate positively with growth at metropolitan scales but turn negative at larger regional or national scales, consistent with the idea that at broader scales coherence may signal path dependence and lock-in rather than productive focus ([straccamore2025?](#)). This core disagreement with other relevant studies ([quatraro2010?](#); [pugliese2019?](#)) raises another challenge as it creates a practical problem for regional diversification analysis: if coherence behaves differently across scales, we need a measure that keeps coherence informative at the regional level without letting it collapse into a purely macro “lock-in” indicator.

Our solution is to bridge the scale disagreement by measuring coherence for regions at a smaller internal scale: instead of computing coherence as a single region-wide property, we compute coherence within broad technology domains (IPC sections, i.e. 1st digit code). The intuition is simple: regional portfolios can be coherent in some domains and not others, and those within-domain alignments are precisely what should matter for whether a region can realistically leverage a technology’s prerequisites and downstream connections in FITS. This is also consistent with innovation geography’s emphasis on “cognitive proximity” and the idea that relatedness works through structured overlaps in knowledge bases rather than mere activity counts (Neffke, Henning, and Boschma 2011; Boschma 2015).

To illustrate this further, let’s consider two regions, both lacking specialization in technol-

ogy t , both having similar potential $p_{r,t,y}$. However, Region A specializes in technologies that are strong predictors of t (strong weighted links), while Region B specializes in technologies unrelated to t in the network (weak weighted links). Coherence captures this difference: if region A has high coherence with t then its portfolio aligns with t 's network prerequisites. Similarly region B's portfolio is misaligned with t thus it would have low coherence.

This is directly aligned with the coherent diversification logic: what matters is not only how many technologies a region have, but whether its technology portfolio forms a coherent capability block at industry level around the target technology (**pugliese2019?**; **quatraro2010?**). The coherence metric also operationalizes the “knowledge coherence” and “cognitive proximity” concepts from innovation literature (Neffke, Henning, and Boschma 2011; Boschma 2015) using our directional network structure. It enables testing whether regional technological potential (predicted feasibility) depends on the alignment in knowledge infrastructure between regional portfolios and technology network positions.

For each region r , technology i , and IPC category c , we construct two embeddedness vectors capturing i 's directional network position and compare them to r 's average embeddedness for technologies in category c .

Technology embeddedness (individual technology i):

- Incoming: $\text{embc_to}_{i,c} = \frac{\sum_{t \in c} W_{t \rightarrow i}}{|\{t \in c: W_{t \rightarrow i} > 0\}|}$ (average FITS weight from category c to technology i)
- Outgoing: $\text{embc_from}_{i,c} = \frac{\sum_{t' \in c} W_{i \rightarrow t'}}{|\{t' \in c: W_{i \rightarrow t'} > 0\}|}$ (average FITS weight from technology i to category c)

Regional average embeddedness (region r , category c):

- Incoming: $\overline{\text{embcat_to}}_{r,c} = \frac{1}{|S_{r,c}|} \sum_{t \in S_{r,c}} \text{embcat_to}_{t,c}$ where $S_{r,c} = \{t \in c : \text{RTA}_{r,t,y} \geq 1\}$
- Outgoing: $\overline{\text{embcat_from}}_{r,c} = \frac{1}{|S_{r,c}|} \sum_{t \in S_{r,c}} \text{embcat_from}_{t,c}$

Coherence is the cosine similarity between technology i 's directional embeddedness and region r 's average directional embeddedness for all categories.

$$\text{Coherence}_{r,i} = \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \cdot \|\mathbf{v}_2\|}$$

where:

- $\mathbf{v}_1 = [\text{embcat_to}_{i,c}, \overline{\text{embcat_to}}_{r,c}]$
- $\mathbf{v}_2 = [\text{embcat_from}_{i,c}, \overline{\text{embcat_from}}_{r,c}]$

Conceptually, this is close to how coherent diversification measures compute “how surrounded a technology is” by related portfolio elements, but with two key differences. First, we ground proximity in predictive, directed dependencies (FITS) rather than symmetric co-occurrence similarity. Second, we compute coherence at an internal domain scale (IPC section) to keep the measure meaningful for regional analysis under known scale heterogeneity.

In here, our coherence measure ranges from 0 to 1. In our context, high coherence (≈ 1) means that for technology i 's embeddedness signal in different categories/industries closely matches the average industrial embeddedness of region r 's portfolio of technologies. The

region’s existing capabilities align with the structural prerequisites and consequences of technology i . Alternatively, a neutral coherence (≈ 0) reflects a misalignment in industrial embeddedness between technology i ’s and the region.

Coherence serves two roles in this work. First, as dyadic fit measure: it is a direct indicator of portfolio–technology alignment in a given domain or industry, capturing whether predicted feasibility is supported by an internally coherent knowledge infrastructure. Second, as moderation channel: it allows us to test whether the effect of technology network position (e.g., bridging technologies, enabling technologies) depends on whether regions have the industry-level knowledge infrastructure to exploit those positions. This is an explicitly contextual mechanism that simple diversification counts cannot reveal (quatraro2010?; Boschma 2015) and aligns with core idea of this work.

3.7 Summary

Our approach replaces traditional relatedness constructs with machine learning-derived measures that enable contextualized diversification analysis:

- **Technological Potential** ($p_{r,t,y}$): Region-specific, non-linear, time-varying probabilities replace linear relatedness density
- **FITS Network** ($W_{t \rightarrow t'}$): Asymmetric, predictive dependencies replace symmetric co-occurrence-based relatedness

- **Coherence** ($\text{Coherence}_{r,t,c,y}$): Structural alignment between regional portfolios and technology network positions captures knowledge infrastructure quality

Together, these measures allow testing how diversification is contingent on regional knowledge infrastructure (RQ2), national ecosystem characteristics (RQ3), and spatial factors (RQ4) in ways that traditional relatedness frameworks cannot—addressing the core problem of contextualizing diversification strategies beyond path dependency identification.

4 Empirical Design

In the preceding chapters we discussed in more details the need for contextualised understanding of regional diversification and we argued that such extra context should account both endogenous and exogenous factors. Analogous to the research agenda in the EEG literature we highlighted that this contextualisation is mainly dependent of three channels: relatedness, network structures, and institutions. In the previous chapter we went a step further and explained why we actually need to go beyond the classical proximity and relatedness paradigm since relatedness and network structure are almost identical constructs. The idea stems from their conceptual and methodological limitations which we summarised in three points: (1) symmetric co-occurrence measures obscure directional technological dependencies, (2) linear aggregation of relatedness density discards combinatorial information about how capabilities interact, and (3) co-occurrence measures yield noisy proximity metric because what's measured almost always have higher dimensions than what is observed. Because of these issues, we proposed a machine learning training strategy that consider relatedness as predictive exercise where we use random forest to estimate alternative measures of proximity: regional technological potential our analogue

to relatedness density, and the Feature Importance Technology Space (FITS) our analogue to the classical Technology Space (and relatedness). Additionally, we also defined a coherence measure following intuition in the literature that we previously discussed. In this chapter we provide an empirical application to our methodological contribution following the same ideas we discussed in the literature review. In here we will first re-iterate and further clarify our objectives and how they translate to empirical constructs, then we will explain progressively how we will to address the research questions and the test the hypothesis.

4.1 Overview and Identification Goals

The central question of this chapter is what explains variation in regional technological potential? The analogue to this question in the REC literature is what explains relatedness. The reason this question is asked at all stems from a large body of studies that show that relatedness density, is a major predictor of entry to new technologies. The more related a region is to a technology the higher the probability of future entry. Similarly, relatedness density is also a major predictor of economic complexity. This is to say that the technological fabric of a region is extremely correlated with growth. In contrast, C. Pinheiro (2025) and Andreoni and Chang (2019) mentioned a structural gap in the way these ideas were addressed in the literature. Specifically, that conventional proximity assessment primarily provide a supply-side representation of what regions might do given their existing portfolios, while remaining comparatively silent on when and where (Hidalgo 2023) those feasibility conditions are strengthened or weakened by network structure, institutional

environments, and spatial interdependence.

Moreover, because our contribution is analogue to relatedness we find it sensible to ask the same question again and extend our methodological contribution to an empirical one as well. We follow the agenda we discussed in the literature review: A contextualised assessment of diversification possibilities through a more contextualised proximity measure (potential), network structure (FITS), institution (Entrepreneurial Ecosystem), and spatial dependence.

To explore this question within the framework of this work, we provide two complementary designs through three modeling approaches each target a specific area of the question. The first empirical design is dyadic, which means it addresses the region-technology as a dyadic structure. In the dyadic design we assess how the position of technologies in the FITS network influence regional technological potential, and whether this influence is universal or dependent on the regional industrial structure, where this dependence is operationalised by our coherence measure. We then extend this by assessing how the inclusion of the entrepreneurial ecosystem covariates affect these relationships. The second design addresses spatial dependence, where we aggregate our covariate into a regional panel and assess whether geographical proximity provide and additional sense to our initial question. As a secondary benchmark, we also compare the influence of potential and relatedness as contrasting construct on regional entry into new technologies.

4.2 Research Design

The guiding research question is whether and how regional portfolio structure mediates the relationship between a technology’s position in the knowledge network and the feasibility of diversification. This decomposes into four testable claims.

- H1 (Benchmark): Regional Technological Potential predicts technology entry better than relatedness density. The literature has established relatedness density as a robust predictor of entry (Hidalgo and Hausmann 2009; Neffke, Henning, and Boschma 2011). We test whether our capability-based potential measure subsumes, supplements, or displaces this predictive power. This hypothesis is not causal; it is a precision benchmark that motivates the subsequent decomposition.
- H2 (Portfolio–network interaction): The effect of a technology’s position in the FITS network on its feasibility is moderated by the alignment between that technology and the regional portfolio. This is the core theoretical claim of the chapter. It draws on two complementary mechanisms: (a) the recombination logic common to evolutionary economic geography, which holds that regions enter new activities by recombining existing capabilities (Frenken and Boschma 2007; Antonietti and Montresor 2021). And (b) the differential value of network links depending on whether the focal activity is related or unrelated to the regional base. On this second point, Zhu, Guo, and He (2021) show that unrelated industries — those peripheral to the regional portfolio — perform better when they can access a small number of strong links rather than a diffuse network of weak connections, because strong links pro-

vide efficient, predictable knowledge transfer under conditions of high uncertainty. We apply an analogous logic at the technology level: a bridging technology (high betweenness) provides a strong, concentrated link between the region’s existing portfolio and a new domain, and we expect this link to be most valuable precisely for regions whose portfolios are *not* yet well-aligned with the target — the asymmetric moderation that constitutes H2b.

- H2 (Portfolio–network interaction): The effect of a technology’s position in the FITS network on feasibility is moderated by portfolio alignment (coherence). This is the core structural claim. It draws on the recombination logic in evolutionary economic geography: regions diversify by recombining existing capabilities, and feasibility depends on whether the regional portfolio provides the relevant interfaces and complementarities needed to support entry (**frenkenboschma2007?**; Boschma 2017). Relatedness therefore captures an important but incomplete place-dependent dimension of diversification (Antonietti and Montresor 2021). FITS extends this logic by distinguishing where a technology sits in the directed capability space—i.e., what kinds of capability pathways it opens or requires—while coherence captures the alignment/consistency of the regional portfolio as a capability system.

We structure H2 around two archetypal network roles that imply different alignment requirements following Antonietti and Montresor (2021) and Montresor and Quatraro (2017):

- Key Enabling Technologies-KET (eigenvector centrality): Technologies that occupy authority-like positions—highly connected to other well-connected technologies—

represent capability-dense domains that tend to require broad, complementary competence. In our interpretation, eigenvector centrality captures this KET-like “deep capability” character: these technologies sit in the structural core and are entangled in many interdependencies. Entry should therefore be more feasible when the regional portfolio is coherent and aligned with the relevant capability bundle.

- General Purpose Technologies-GPT (betweenness centrality): Technologies with high betweenness sit on many shortest directed paths and act as stepping stones between otherwise distant domains. In our interpretation, betweenness captures GPT-like “bridging capability” roles: cross-cutting technologies that connect clusters and enable recombination across domains. Their value as entry points should be less dependent on prior alignment, because their function is precisely to provide a pathway into adjacent capability space.

We therefore test the following sub-claims:

- H2a (Alignment): Coherence positively predicts potential. More coherent portfolios provide complementary infrastructures that reduce feasibility barriers for entry.
- H2b (GPT-like moderation): The positive association between betweenness centrality and potential is weaker under high coherence (or equivalently, stronger when coherence is lower). Bridge/GPT-like technologies should matter most when direct alignment is limited, because they provide targeted pathways into new capability domains.
- H2c (KET-like moderation): The association between eigenvector centrality and

potential is more sensitive to coherence than the betweenness effect. If eigenvector-central technologies reflect KET-like capability cores, then feasibility should increase disproportionately with portfolio alignment, because entry requires a coherent and complementary base.

- H2d (Distance): Topological distance from the regional frontier compounds feasibility barriers. Where a region must traverse more directed capability steps in FITS to reach a target technology, feasibility should decline, and this effect may be especially pronounced for core/KET-like targets.
- H3 (National ecosystem): National entrepreneurial ecosystem characteristics shape regional technological potential beyond dyadic portfolio mechanisms. Relatedness and portfolio configuration describe endogenous opportunity structure, but capability mobilisation depends on enabling conditions—finance, competition, innovation capacity, and institutional bottlenecks—that operate at least partly at the national level (Acs, Autio, and Szerb 2014; C. Pinheiro 2025). We therefore test whether national ecosystem measures explain variation in regional potential after accounting for portfolio structure and technology-network mechanisms.
- H4 (Spatial dependence): Regional potential exhibits positive spatial dependence beyond what regional characteristics explain. If high-potential regions systematically neighbour other high-potential regions after conditioning on observed portfolio and institutional characteristics, this indicates genuine interregional interdependence—knowledge spillovers, demonstration effects, or co-development of capabilities across borders—rather than merely correlated endowments (Boschma

Hypotheses, Variables, and Empirical Tests

Hypothesis	Claim	Key variables	Where tested
H1	Potential outperforms relatedness density in predicting entry	pred11 vs relatedness_density	Table 5.1
H2a	Coherence positively predicts potential	coherence	Tables 5.2-5.3
H2b	Betweenness benefit stronger for low-coherence regions	betweenness x coherence	Table 5.3
H2c	Eigenvector penalty independent of coherence	eigenvector x coherence	Table 5.3
H2d	Distance compounds eigenvector difficulty	rawdist x eigenvector	Table 5.3
H3	National ecosystem shapes potential beyond dyadic mechanisms	GEI pillars x8-x14	Tables 5.4-5.5
H4	Positive spatial dependence in regional potential	Spatial lag (SAR)	Tables 5.6-5.7

Note: H2a-H2d are all estimated in the dyadic fixed effects framework (Models 1-3). FE = two-way fixed effects (region + year); RE = hierarchical random effects (regions nested within countries); SAR = spatial autoregressive panel model.

and Iammarino 2009). We test this by moving from dyadic modelling to region-level panels that can accommodate spatial dependence in feasibility.

4.3 Data

We use data very similar to many studies discussed in the literature review, namely the European patent applications from the EPO/PATSTAT database, covering 641 technologies (4-digit IPC classes) across 345 NUTS2 regions in 30 European countries over 11 years (2008–2018). The dyadic models (H1–H3) use approximately 1.9 million technology–region–year observations. The spatial models (H4) aggregate to approximately 3,200 observations. Patent data serve as a standard proxy for regional knowledge production and technological specialisation. We briefly described in the previous chapter how we use this

data to construct our training strategy and build our empirical constructs (Regional Technological Potential, and the Feature Importance Technology Space (FITS)). In here we take this a step further and describe in more details the other variables we use in our design.

4.3.1 Outcome: Regional Technological Potential

The primary outcome in this work is Regional Technological Potential $p_{r,t,y}$: the Random Forest-predicted probability that region r develops revealed technological advantage (RTA 1) in technology t at year y , estimated from portfolio characteristics four years prior. Values are bounded between 0 and 1.

Because OLS on a bounded outcome is problematic and can generate out-of-range fitted values, we apply a logit transformation before estimation:

$$p_{r,t,y}^* = \text{logit}(p_{r,t,y}) = \log\left(\frac{p_{r,t,y}}{1 - p_{r,t,y}}\right)$$

with extreme values trimmed to avoid numerical instability. The transformed outcome $p_{r,t,y}^*$ is then unbounded and appropriate for OLS estimation. Ideally, a beta or a fractional logit regression would be more appropriate given the bounded nature of the outcome. We did test these approaches during the modeling process and we decided that given the size of the data and the complexity of the specifications the computational cost is great and numerical stability becomes an issue. Additionally, our decision to decision to use OLS estimation is also motivated by the need for a clear interpretation of the results.

For the entry benchmark (H1), we use a standardised version of the untransformed potential, $\tilde{p}_{r,t,y}$ (z-scored), as a predictor of next-period entry alongside standard relatedness density (z-scored).

For the spatial models (H4), we aggregate to region–year means:

$$P_{r,y} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} p_{r,t,y}$$

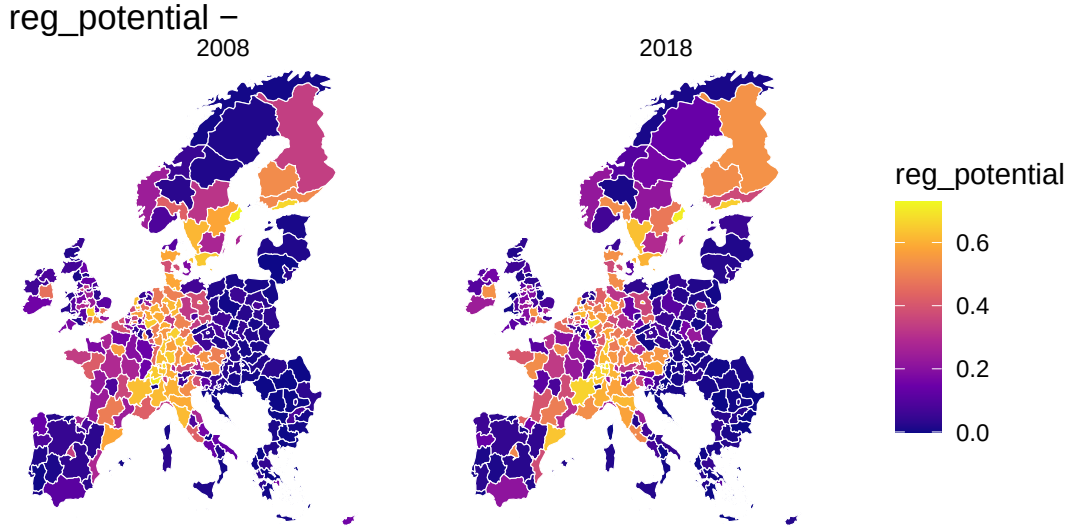


Figure 4.1

4.3.2 Technology–Region Level (Dyadic) Variables

Coherence ($\text{coherence}_{r,t,y}$) measures the alignment between the regional technology portfolio and a specific technology in the directed FITS network in terms of industrial embeddedness (see the previous chapter for more details). It captures whether the regional

portfolio is structurally positioned across industries to support a given technology’s embeddedness in those industries. i.e., whether the region and the technology “sit” in compatible parts of the capability system. This aligns with the intuition in the literature that diversification feasibility depends on how well the target technology aligns with internal regional knowledge infrastructure base (teece1994?; nesta2005?; quatraro2010?; pugliese2019coherent?; straccamore2025fitnesscoherence?).

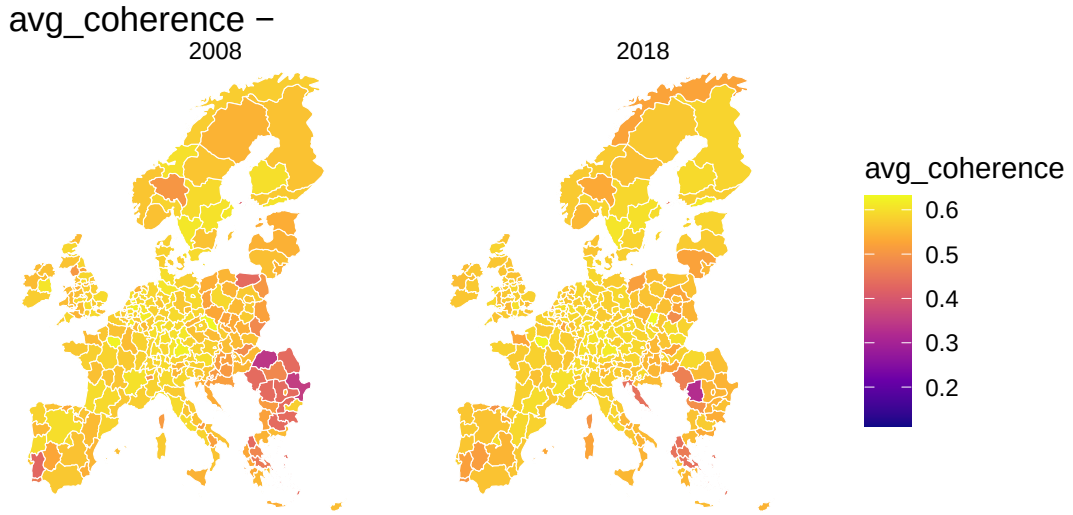


Figure 4.2

Distance ($\text{rawd}_{r,t,y}$) is the shortest path weighted length in the FITS network from the closest technology in the regional portfolio to a given technology regardless of the direction of the links. This distance measures topological accessibility in the network or how many steps separate the region’s current frontier from a given opportunity. This variable operationalises the product-space logic of Hidalgo et al. (2007) who show that the probability of entry decays with distance in the product space.

Patent count ($\text{patents}_{r,t,y}$) is the number of patent applications attributed to region r in technology t at year y . We use it to controls for pre-existing knowledge stock on a given technology in the region. This is to ensure that coherence and network position effects are do not turn into proxies for size as argued in (batheltstorper2023?).

4.3.3 Technology Level Variables

Betweenness centrality¹ (betweenness_t) measures how frequently a technology lies on shortest directed paths between other technologies. High-betweenness technologies act as bridges between otherwise separated domains of knowledge. In our interpretation, this is the structural analogue of GPT-like enabling roles where technologies that facilitate transitions across domains serve as an intermediary (Antonietti and Montresor 2021).

Eigenvector centrality (eigenvector_t) measures how strongly connected a technology is to other highly connected technologies. The idea is to capture authoritative positions in the capability system. In our interpretation, this corresponds to KET-like roles capability-dense, strongly interdependent domains whose entry tends to require broader complementary competence rather than a single bridge.

¹

Regional complexity ($\text{reg_comp}_{r,y}$) and **technology complexity** (tech_comp_t) are included as controls following the economic complexity tradition in which diversity–ubiquity

¹the FITS network is directed, in earlier iterations explored HITS authority and hub scores separately but because HITS scores and betweenness are correlated we decided to use eigenvector centrality in the final specification.

Table 4.1

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
year	11	0	2013.00	3.16	2008.00	2013.00	2018.00
pred1	1619198	0	0.18	0.21	0.00	0.09	1.00
patent_n	593	0	1.20	12.59	0.00	0.00	2377.00
coherence	1985135	0	0.55	0.22	0.00	0.57	1.00
betweenness	3156	0	0.00	0.00	0.00	0.00	0.05
authority	7039	0	0.34	0.19	0.00	0.29	1.00
hub	5529	0	0.06	0.12	0.00	0.01	1.00
degree_in	7046	0	0.65	0.16	0.00	0.64	1.00
degree_out	5534	0	0.11	0.18	0.00	0.02	1.00
degree_total	7047	0	0.20	0.15	0.00	0.13	1.00
eigenvector	7051	0	0.10	0.16	0.00	0.04	1.00
rawdist	5637	4	132.61	242.27	26.46	90.94	2883.96
relatedness_density	5537	4	9.42	8.04	0.00	7.81	100.00
reg_comp	3087	4	-0.00	1.00	-12.98	0.08	5.44
tech_comp	6833	4	-0.00	1.00	-23.15	0.01	4.63
entry	3	18	0.06	0.23	0.00	0.00	1.00

patterns proxy underlying capability endowments (**hidalgohausmann2009?**). This mirrors the rationale for including regional complexity controls in related European diversification work (Antonietti and Montresor 2021).

4.3.4 National Level Variables: GEI Pillars

To test H3, which assumes that national entrepreneurial ecosystem characteristics contribute to regional potential, we incorporate seven pillars from the Global Entrepreneurship Index (GEI). The GEI decomposes national ecosystem characteristics into fourteen that account for individual attitudes, systemic capabilities, and aspirational orientations (Acs, Autio, and Szerb 2014).

For the purposes of this work, we retain pillars x8–x14. The reason here is two folds, first

Table 4.2

	pred1	rawdist	patent_n	coherence	betweenness	eigenvector	authority
pred1	1	.	.	.	.	.	.
rawdist	-.07	1	.	.	.	.	.
patent_n	.24	-.02	1	.	.	.	.
coherence	.40	-.19	.11	1	.	.	.
betweenness	.23	-.10	.05	.40	1	.	.
eigenvector	-.15	.08	-.02	-.21	-.17	1	.
authority	-.13	-.03	-.06	-.18	-.25	-.06	1
hub	.20	-.06	.12	.33	.15	-.06	.10
relatedness_density	.69	-.00	.22	.12	.02	-.02	-.01
reg_comp	.34	.00	.09	.03	-.01	.01	.01
tech_comp	-.20	.05	-.11	-.28	-.24	-.01	-.02
entry	.29	-.02	.26	.13	.07	-.05	-.05

pillars x1-x6 capture individual-level entrepreneurial attitudes² and mostly reflect social preferences of the entrepreneurial ecosystem and not the institutional landscape.

Second, pillars x1–x6 are heavily correlated ($r = .57-.88$ across all pairwise combinations, as shown in Table 4.3). This is problematic in our case since it will create multicollinearity issues in the models and would compromise numerical stability, fit, and most likely bias the results.

Thus, we retained pillars x8-x14 that show higher heterogeneity and more moderate pairwise correlations. Regardless, the pillars we retain (see Table 4.4) are more informative for our case since they capture the institutional landscape of entrepreneurial ecosystem which is our main area of focus in this work.

Because GEI pillars are measured at the national level and vary primarily across countries rather than within countries over time, they cannot be identified in a two-way region-and-

²opportunity perception, startup skills, risk acceptance, networking, cultural support, opportunity startup

Table 4.3

	x1_opportunity_perception	x2_startup_skills	x3_risk_acceptance
x1_opportunity_perception	1	.	.
x2_startup_skills	.33	1	.
x3_risk_acceptance	.68	.41	1
x4_networking	.71	.33	.57
x5_cultural_support	.88	.36	.73
x6_opportunity_startup	.85	.41	.72
x7_technology_absorption	.58	.21	.56
x8_human_capital	.46	.17	.37
x9_competition	.76	.21	.66
x10_product_innovation	.59	.18	.45
x11_process_innovation	.60	.33	.63
x12_high_growth	.51	.07	.33
x13_internationalization	.01	-.15	.06
x14_risk_capital	.46	.04	.44

Table 4.4

GEI Pillars Retained in Analysis

Pillar	Variable name	Role in capability development
Human capital	x8_human_capital	Education and training infrastructure for technology development
Competition	x9_competition	Market pressure driving diversification and upgrading
Product innovation	x10_product_innovation	National capacity for new product development
Process innovation	x11_process_innovation	Technology absorption and adaptation capacity
High growth	x12_high_growth	Scaling orientation enabling specialisation
Internationalisation	x13_internationalization	Export orientation and global knowledge access
Risk capital	x14_risk_capital	Venture financing enabling exploration of new technologies

Table 4.5

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
year	10	0	2013.00	3.01	2008.00	2013.00	2018.00
x1_opportunity_perception	185	0	0.55	0.27	0.13	0.50	1.00
x2_startup_skills	200	0	0.62	0.20	0.15	0.62	1.00
x3_risk_acceptance	187	0	0.61	0.26	0.10	0.66	1.00
x4_networking	193	0	0.50	0.21	0.13	0.45	1.00
x5_cultural_support	180	0	0.60	0.26	0.09	0.58	1.00
x6_opportunity_startup	181	0	0.66	0.24	0.15	0.62	1.00
x7_technology_absorption	187	0	0.66	0.22	0.14	0.66	1.00
x8_human_capital	199	0	0.51	0.20	0.14	0.48	1.00
x9_competition	191	0	0.59	0.26	0.17	0.56	1.00
x10_product_innovation	195	0	0.55	0.24	0.06	0.56	1.00
x11_process_innovation	198	0	0.63	0.20	0.22	0.64	1.00
x12_high_growth	204	0	0.51	0.17	0.12	0.52	0.95
x13_internationalization	185	0	0.69	0.20	0.23	0.72	1.00
x14_risk_capital	192	0	0.63	0.20	0.19	0.62	1.00

year fixed effects specification — national variation would be absorbed by region fixed effects. This is why H3 is tested in two complementary ways: (i) in fixed effects models, and (ii) in hierarchical random effects models where are regions nested within countries, where GEI variables operate at the country level as intended.

4.3.5 Region Level Variables (Spatial Models)

For the spatial panel (H4), we aggregate the dyadic variables to region–year means. Specifically we retain average coherence, average betweenness, and average eigenvector of each region’s existing portfolio. Additionally, we also aggregate patent counts to the total regional patenting activity across all technologies and keep regional complexity. Each of these variables reflect an aggregate characteristic of the regional technological portfolio. Further descriptive details on these variables are presented in Table 4.6 and Table 4.7.

Table 4.6

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
year	11	0	2013.00	3.16	2008.00	2013.00	2018.00
reg_potential	1565	0	0.22	0.23	0.00	0.12	0.76
patent_count	1249	0	776.19	1985.54	0.00	186.00	27222.00
n_techs	368	0	90.03	92.69	0.00	59.00	496.00
n_specialized	210	0	59.81	49.01	0.00	50.00	241.00
diversity0	367	0	3.70	1.58	0.00	4.08	6.21
diversity	2851	0	3.21	1.33	0.00	3.58	5.28
avg_coherence	3097	0	0.55	0.04	0.11	0.56	0.64
avg_tech_comp	11	0	-0.00	0.00	-0.00	0.00	0.00
entry_rate	1946	0	0.06	0.05	0.00	0.05	0.26
avg_rd	3083	0	9.42	7.74	0.00	7.97	38.71
avg_dist	11	0	132.61	13.50	107.08	137.50	151.91
avg_betweenness	3080	0	0.00	0.00	0.00	0.00	0.04
avg_eigenvector	3087	0	0.07	0.04	0.00	0.06	0.52
med_betweenness_spec	2106	0	0.00	0.00	0.00	0.00	0.04
med_eigenvector_spec	2554	0	0.03	0.03	0.00	0.02	0.52
med_betweenness_exist	1925	0	0.00	0.00	0.00	0.00	0.04
med_eigenvector_exist	2455	0	0.03	0.03	0.00	0.02	0.52
pct_high_between_spec	124	0	0.46	0.12	0.00	0.50	0.50
pct_high_eigen_spec	109	0	0.46	0.12	0.00	0.50	0.50
pct_high_between_exist	197	0	0.46	0.12	0.00	0.50	0.50
pct_high_eigen_exist	183	0	0.46	0.12	0.00	0.50	0.50

Table 4.7

	reg_potential	patent_count	n_techs	diversity	avg_coherence	avg_betweenness
reg_potential	1	.	.	.	.	.
patent_count	.51	1	.	.	.	.
n_techs	.91	.75	1	.	.	.
diversity	.74	.37	.74	1	.	.
avg_coherence	.52	.30	.54	.77	1	.
avg_betweenness	-.05	-.09	-.06	.13	.20	1
avg_eigenvector	.20	.17	.23	.27	.28	.28
entry_rate	.76	.39	.78	.78	.52	.52
avg_rd	.91	.58	.95	.85	.59	.59

4.4 Model 1: Dyadic Fixed Effects Models

The first model in our work is specified to investigate the explanatory factors that drives regional technological potential. In here, as well as in the other dyadic models we estimate our linear models on the logit-transformed potential $p_{r,t,y}^*$. Essentially we adopt a two-way region and year fixed effects as our preferred specification to assess the within region variations with standard errors clustered by region and technology. This choice accounts for the structural correlation of the observations in the same region or technology.

The general specification is:

$$p_{r,t,y}^* = \alpha_r + \alpha_y + \mathbf{X}_{r,t,y}\beta + \varepsilon_{r,t,y}$$

where α_r are region fixed effects, α_y are year fixed effects, and $\mathbf{X}_{r,t,y}$ is the covariates matrix.

This specification is one part of our identification strategy where region fixed effects absorb all time-invariant regional characteristics that could confound the relationship between the regional portfolio characteristics and regional potential. Additionally, year fixed effects absorb all common temporal shocks. Thus, we identify within-group variations that change over time. Additionally, this specification should also address at least partially potential issues with endogeneity.

The idea here is to understand what factors explain regional technological potential when we account for the regional and temporal common variations. Does coherence moderate

the effect of network position on potential? Is this moderation identical for KETs and GPTs?

Regardless, we estimate a multitude of regression models and progressively tighten their complexity and specifications. The idea here is to compare the consistency of the contribution of the covariates across these specifications.

The baseline model includes coherence, distance, patent count, and complexity controls. The network model adds betweenness and eigenvector centrality testing whether network position consistently predicts potential. The interaction model add the coherence x betweenness and coherence x eigenvector terms. We also add a parallel specification for robustness where we replaces the coherence moderation with distance.

Of course with the inclusion of national ecosystem covariates, this identification looses some of its merits since we cannot account for cross-regional country level differences. This is because region fixed effect will absorb country level variations and because country fixed effect will leave out common regional variation. This motivates the adoption of a hierarchical model when we include the country level covariates.

4.5 Model 2: Hierarchical Models

As we cross different literature domains in this work we observed that there's a clear and urgent gap that was not properly addressed that's the lack of time varying regional entrepreneurial ecosystem datasets. The Regional Entrepreneurship and Development Index (REDI) (Szerb et al. 2013; Argiles et al. 2014) and the work of Leendertse, Schrijvers, and

Stam (2022) are a step in that direction but the challenge of time variation still persist. This means that to be consistent with the evolutionary scope with which we framed this work we are forced to accommodate the national covariates.

Although the fixed effect design we described in the previous section is relatively adequate for isolating the within-region mechanisms it still does not draw the full story as it discards the between-region variations. Yet between-region variation in our context is theoretically meaningful since we are interested in understanding why some regions are systematically characterised by higher potential compared to others. This interest then compliments our focus in the previous specification where the focal attention is on why potential fluctuates.

Thus to properly test H3, we need a specification that preserves nation and regional variation. To this end we include random effects by modelling nested region within country intercepts that are drawn from a distribution rather than absorbing them as free parameters. For that we estimate linear mixed models with the following nesting structure:

$$p_{r,t,y}^* = \mathbf{X}_{r,t,y}\beta + \zeta_t + u_c + u_{r|c} + \varepsilon_{r,t,y}$$

where $u_c \sim \mathcal{N}(0, \sigma_c^2)$ is a country-level random intercept and $u_{r|c} \sim \mathcal{N}(0, \sigma_{r|c}^2)$ is a region-within-country random intercept. Year is included as a dummy (ζ_t) to absorb common temporal dynamics.

With these models we will be able to compare the contribution of the covariates to the contribution to the nested structure itself. This is to say that if the nested structure is

highly predictive of the variation in regional technological potential then the outcome is primarily driven by the location rather than the covariates themselves. This justifies our next layer of analysis: if geography matters then we need to assess the spatial dependence of the factors associated with regional potential. To that end we explore in the next section in more details how we will assess this dependence.

4.6 Model 3: Spatial Panel Models

Till now, we discussed two models that investigate variation in regional technological potential at different levels of grouping specification. The fixed effect models are designed to assess the within-region variation, and the hierarchical model is designed to simultaneously account for the national ecosystem factors while also serving as a robustness check for the network covariates and their interaction with coherence. As we justified earlier, if the nested structure of the data drives much of the explanatory power of the model, it could be an indication that spatial dependence and/or geographical proximity is what matters here. Additionally, the literature gives reason to expect real interdependence especially in the context of European regions where administrative borders are simply a construct and not a barrier for firms and entrepreneurs (Boschma and Iammarino 2009; Komlósi et al. 2025). The spatial models in this analysis test this formally.

To proceed with the spatial analysis, we construct spatial weights using k-nearest neighbours ($k = 5$) based on NUTS2 geographic centroids. We also perform row-standardisation for each focal region i and its candidate neighbour j , such that $\sum_j w_{ij} = 1$, where:

Table 4.8

Table 3. Spatial Autocorrelation Diagnostics

Baltagi-Song-Koh robust LM tests for panel spatial dependence; $k = 5$ nearest-neighbour weights

Test	Statistic	p-value	Interpretation
Moran's I (range across years)	0.44-0.51	< 0.001 (all years)	Strong positive spatial autocorrelation in regional potential
LM-lag (standard)	-	< 0.001	Spatial lag process significant before conditioning
LM-error (standard)	-	< 0.001	Spatial error process significant before conditioning
Robust LM-lag	350.4	< 0.001	Spatial lag dominates after conditioning error
Robust LM-error	4.1	0.042	Marginal - spatial error secondary to lag process

Note: Robust LM tests condition each null on the presence of the other type of dependence (Baltagi, Song, and Koh 2003). The diagnostic pattern - highly significant robust LM-lag, marginally significant robust LM-error - favours the SAR specification.

$$w_{ij} = \begin{cases} 1/k & \text{if } j \in \text{knn}_5(i) \\ 0 & \text{otherwise} \end{cases}$$

The preliminary diagnostics are summarised in Table 4.8 show that Moran's I statistics for our main outcome reveal a persistent positive spatial autocorrelation across all years in the panel (I around 0.44-0.51, $p < 0.001$). Furthermore, we apply the Baltagi-Song-Koh Lagrange Multiplier tests designed for panel data to discriminate between competing spatial mechanisms. The robust LM-lag statistic is large and highly significant (350.4, $p < 0.001$), while the robust LM-error statistic is only marginally significant (4.1, $p = 0.042$) after conditioning on the spatial lag. Thus, these diagnostic results indicate that spatial dependence operates primarily through neighbours' outcomes rather than through

spatially correlated errors. In light of this, the SAR model is the adequate specification, while we use the SEM model as a robustness check.

4.6.1 Spatial Autoregressive (SAR) Panel - Primary Specification

We estimate the Spatial Autoregressive (SAR) model with two-way (region and year) demeaning. In this model, the spatial autoregressive parameter ρ tests H4 where a positive and significant ρ indicates that regions tend to exhibit higher potential when neighbouring regions also display high potential.

$$P_{r,y} = \rho \mathbf{W} P_{r,y} + \mathbf{X}_{r,y} \beta + \alpha_r + \zeta_y + \varepsilon_{r,y}$$

The covariates matrix $\mathbf{X}_{r,y}$ includes regional patent count, regional complexity, average coherence, average betweenness, and average eigenvector, along with their interactions.

However, we need to note here that at the dyadic level, betweenness and eigenvector measures are technology level metrics. They capture the characteristics of the network position of a given technology whether it's a key enabling technology or general purpose technology. In this analysis however, average eigenvector and average betweenness measure how much a region is endowed with key enabling technology or general purpose technology in their portfolio and not the characteristics of the technologies themselves. That is to say in this level of analysis our focus is on the aggregate characteristics of the regional portfolio. For our work, this interpretation complements the specification of the dyadic models and reveal more context to the dynamics we're studying.

4.6.2 Spatial Error Model (SEM) - Robustness

Alternatively, we estimate the Spatial Error Model (SEM) as a robustness check. In this model we assume that the unobservable determinant of regional potential are spatially autocorrelated. This is to say that the spatial dependence of regional potential is driven by unobserved spatial correlation rather than observed variations.

$$P_{r,y} = \mathbf{X}_{r,y}\beta + \alpha_r + u_{r,y}, \quad u_{r,y} = \delta \mathbf{W}u_{r,y} + \varepsilon_{r,y}$$

In here, δ is our benchmark such that its sign, magnitude and significance compared to ρ from the SAR model allows to differentiate the spatial dynamics that drives regional potential.

5 Results

Our empirical analysis proceeds in five stages. First, we benchmark how much standard relatedness density and our RF-based Potential explain next-period entry under progressively stricter comparisons (Table Table 5.1). Second, we examine how technology network positions, regional knowledge infrastructure, and accessibility shape diversification potential using region and year fixed effects (Table Table 5.2). Third, we incorporate national entrepreneurial ecosystem characteristics to test whether these dyadic mechanisms persist once national institutions are accounted for (Table Table 5.3). Fourth, we estimate hierarchical random effects models to reflect the nesting of regions within countries and to assess how much variation lies between versus within regional units (Table Table 5.4). Fifth, we aggregate potential to the regional level and estimate spatial panel models to test whether regional potential exhibits spatial dependence consistent with spillovers or correlated unobservables (Tables Table 5.5 and Table 5.6). Together, these analyses test our hypotheses while checking whether conclusions depend on model structure or level of aggregation.

5.1 Entry

In Table 5.1 we benchmark how much standard relatedness density and our RF-based Potential (pred11) explain next-period entry when we progressively tighten comparisons. In the pooled specifications, both measures strongly predict entry: relatedness density is positive and precisely estimated (0.055), while Potential is larger (0.081). This is consistent with the interpretation that both capture opportunity, while Potential is closer to feasibility as learned from portfolio configurations. Once we move to the fixed-effects designs and absorb region and year fixed effects, interpretation shifts to within-region changes over time, net of common year shocks. In this setting, relatedness density remains strongly positive (e.g., 0.089 in the density-only fixed effects model), and Potential also remains strongly positive (0.089 in the Potential-only fixed effects model), indicating that both track time-varying entry propensity beyond stable regional differences.

The most informative column is the joint fixed effects model, where both predictors enter together. Here, relatedness density drops from 0.089 to 0.028, while Potential remains large at 0.086. This pattern implies that Potential subsumes a substantial share of what density captures, but not all of it: density retains an independent association with entry even after controlling for Potential. In other words, Potential appears to be the dominant predictor in this horse race, while density retains residual explanatory content that may reflect a simpler proximity component that survives beyond the non-linear feasibility signal embedded in Potential. This dominance is also reflected in fit: within- R^2 rises from 0.014 in the density-only fixed effects model to 0.062 in the Potential-only fixed effects model and remains 0.062 in the joint model. This benchmark motivates the subsequent mechanism

analysis: rather than treating relatedness as a sufficient summary statistic, we examine how portfolio alignment, network position, and accessibility structure feasibility. This move is consistent with the broader methodological critique that pairwise relatedness can miss higher-order portfolio information; for example, (**RelatednessMachineLearning?**) argues that shifting from “two-product correlations” to information that captures richer configurations can improve predictive performance. This benchmark motivates our subsequent mechanism analysis: having established that Potential dominates relatedness density in predicting entry, we now decompose Potential itself to examine how portfolio alignment (coherence), network position (betweenness, eigenvector), and accessibility structure feasibility at the dyadic technology-region level

5.2 Network Effects and Coherence Moderation

Table 5.2 presents our core test of how technology network positions and regional knowledge infrastructure shape diversification potential. Column 4, our preferred specification with region and year fixed effects, shows that coherence—the alignment between a technology’s FITS embeddedness and the region’s portfolio—strongly predicts potential (0.706). This aligns with the capabilities-based interpretation of relatedness in evolutionary economic geography. (**TechNetworkResilience?**) states that “economic activities are related through sharing various capabilities,” and it links diversification to learning enabled by “cognitive proximity and interactive learning” following Boschma’s proximity framework. In our setting, coherence operationalizes this shared-capability, portfolio-fit mechanism and remains stable across specifications, consistent with its interpretation as

Table 5.1

	baseline1	baseline2	full1	full2
(Intercept)	0.069*** (0.002)	0.070*** (0.001)		
patent_n	0.008 (0.007)	-0.007** (0.002)	0.017* (0.008)	0.004 (0.003)
relatedness_density	0.055*** (0.002)		0.089*** (0.006)	
reg_comp	-0.003*** (0.001)	-0.003*** (0.001)	-0.001*** (0.000)	-0.000 (0.000)
tech_comp	-0.015*** (0.002)	-0.005*** (0.001)	-0.013*** (0.002)	-0.004*** (0.001)
pred11		0.081*** (0.002)		0.089*** (0.002)
Num.Obs.	1571364	1571364	1571364	1571364
R2	0.043	0.086	0.046	0.092
R2 Adj.	0.043	0.086	0.046	0.092
R2 Within			0.014	0.062
R2 Within Adj.			0.014	0.062
AIC	-36976.1	-109812.6	-41129.6	-119089.6
BIC	-36914.8	-109751.3	-37375.8	-115335.8
RMSE	0.24	0.23	0.24	0.23
Std.Errors	by: region & tech	by: region & tech	by: region & tech	by: region & tech
FE: region			X	X
FE: year			X	X
FE: tech				

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

a structural feasibility correlate rather than a modeling artifact.

Betweenness centrality, capturing bridge positions that connect domains, exhibits substantial positive effects (0.388). However, the coherence $\times$ betweenness interaction is significantly negative (-0.284), revealing contextual contingency: bridging positions disproportionately benefit regions with lower coherence, offering catch-up opportunities when portfolios are misaligned with these bridge technologies. Conversely, high-coherence regions gain less from bridging positions, suggesting that their feasible opportunities are less dependent on bridge roles and more concentrated in already-aligned portions of the technology space. This contingency is conceptually consistent with a tension highlighted in (**TechNetworkResilience?**). After reviewing the standard logic that related variety supports diversification, it notes that interdependencies can also transmit shocks and that evidence on how internal relatedness structures shape outcomes is mixed; it concludes that it is “still unclear” how relatedness shapes resilience and that “networks are of great assistance” because economies are “webs” of interdependent knowledge and activities. Our interaction result is consistent with that network-conditional view: the value of a network position depends on portfolio context rather than being uniform.

The eigenvector centrality pattern presents a contrasting story. Technologies occupying authority positions consistently inhibit entry (e.g., -0.221 in Column 4), yet the coherence $\times$ eigenvector interaction remains non-significant. A conservative interpretation is that authority-like targets are systematically difficult to enter regardless of alignment as captured by coherence. This pattern is compatible with arguments that diversification requires more than proximity. (**GoingBeyondRelatednessKETs?**) argues that diver-

Table 5.2: logit(pred1) - ols

	baseline	network	interaction_a	interaction_a1
(Intercept)	-2.528*** (0.070)		-2.462*** (0.074)	-2.476*** (0.074)
patent_n	0.211*** (0.054)	0.088*** (0.023)	0.213*** (0.053)	0.215*** (0.054)
coherence	1.026*** (0.038)	0.779*** (0.039)	0.898*** (0.039)	0.909*** (0.040)
rawdlist	-0.028 (0.028)	-0.039 (0.028)	-0.011 (0.027)	
reg_comp	0.741*** (0.065)	0.013 (0.018)	0.746*** (0.066)	0.741*** (0.065)
tech_comp	-0.213*** (0.035)	-0.256*** (0.040)	-0.212*** (0.035)	-0.198*** (0.034)
betweenness		0.131*** (0.023)	0.249*** (0.039)	0.266*** (0.040)
eigenvector		-0.237*** (0.030)	-0.206*** (0.028)	-0.201*** (0.029)
betweenness × coherence			-0.184*** (0.032)	-0.191*** (0.033)
coherence × eigenvector			-0.015 (0.030)	-0.001 (0.031)
betweenness × rawdlist				
rawdlist × eigenvector				
Num.Obs.	1906137	1906137	1906137	1922371
R2	0.354	0.659	0.366	0.367
R2 Adj.	0.354	0.659	0.366	0.367
R2 Within		0.356		
R2 Within Adj.		0.356		
AIC	7682716.6	6466221.8	7645396.3	7738488.6
BIC	7682791.4	6470084.6	7645520.9	7738600.8
RMSE	1.82	1.32	1.80	1.81
Std.Errors	by: region & tech	by: region & tech	by: region & tech	by: region & tech
FE: year		X		
FE: region		X		

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

sification is “going beyond relatedness” and reports that enabling technologies support more exploratory trajectories “only when” they are used by other local technologies, implying a complementarity and integration condition beyond portfolio fit alone. Our null moderation for eigenvector is consistent with the idea that barriers for authority targets reflect complements or coordination not captured by alignment measures.

Distance becomes most informative once interactions are introduced. In Table Table 5.2, the main effect of rawdist is not stable in the fixed effects setting, but the interaction between rawdist and eigenvector is negative and robust (around -0.080 in the distance-interaction specifications). This indicates that authority technologies become particularly difficult when they are also topologically remote from any regional foothold. This mechanism is close to the account in (**StrongLinksWeakLinksUnrelatedSurvive?**), which describes network links as “pipelines” and states that “the strength of links reflects technological distance,” with the implication that predictable transfer is more feasible through stronger, closer links. In contrast, the betweenness $\times$ rawdist interaction is small and not robust in the fixed effects specifications (e.g., 0.067 and 0.057 with large standard errors), suggesting limited evidence in within-region comparisons that bridge positions systematically enable longer jumps purely through remoteness.

5.3 National Context and Robustness

Table 5.3 extends our framework by incorporating national entrepreneurial ecosystem characteristics from the Global Entrepreneurship Index. Column 4 shows that the core

dyadic mechanisms persist: coherence remains positive (0.709), betweenness remains positive (0.398), and the coherence $\times$ betweenness interaction remains negative (-0.292). The eigenvector penalty likewise persists (around -0.222) without coherence moderation, and the distance–authority interaction remains negative (rawdist $\times$ eigenvector around -0.082). This stability supports a cautious conclusion that the coherence and network-position mechanisms are not simply proxies for national institutional variation. At the same time, national ecosystem variables contribute additional explanatory structure. Competition is consistently positive (about 0.088 in Column 4), while internationalization is consistently negative (about -0.082). Other GEI components show heterogeneous patterns across specifications, suggesting they may operate differently in cross-sectional comparisons than in within-country temporal change. This combination of stable dyadic mechanisms and non-negligible national effects is consistent with **(RelatednessAndEconComplexityPolicy?)**, which cautions against treating relatedness metrics as sufficient guides to diversification because institutional and policy context can shape which opportunities are realized. This heterogeneity notwithstanding, the consistency of competition and internationalization effects, alongside the stable core mechanisms, provides sufficient evidence for H3 that national entrepreneurial ecosystems shape regional potential beyond dyadic technology-region characteristics.

5.4 Hierarchical Structure and Spatial Implications

Table 5.4 Table 5.4 employs random effects models that account for regions nested within countries, a specification that matches the hierarchical structure of the data and is appro-

Table 5.3: logit(pred1) - ols

	h0	h01	h1	h2
(Intercept)	-2.493*** (0.060)		-2.406*** (0.063)	
patent_n	0.181*** (0.044)	0.090*** (0.023)	0.181*** (0.045)	0.090*** (0.024)
coherence	0.884*** (0.039)	0.784*** (0.040)	0.826*** (0.038)	0.709*** (0.038)
rawdist	-0.024 (0.030)	-0.037 (0.030)	-0.015 (0.029)	-0.027 (0.029)
reg_comp	0.519*** (0.063)	0.015 (0.021)	0.518*** (0.063)	0.011 (0.020)
tech_comp	-0.215*** (0.039)	-0.243*** (0.042)	-0.214*** (0.039)	-0.241*** (0.041)
betweenness	0.104*** (0.023)	0.133*** (0.024)	0.316*** (0.038)	0.398*** (0.038)
eigenvector	-0.224*** (0.031)	-0.240*** (0.031)	-0.211*** (0.028)	-0.222*** (0.028)
x8_human_capital	-0.132* (0.054)	0.044* (0.022)	-0.134* (0.054)	0.044* (0.022)
x9_competition	0.568*** (0.082)	0.087* (0.034)	0.573*** (0.082)	0.088** (0.034)
x10_product_innovation	0.198*** (0.056)	0.016 (0.018)	0.200*** (0.056)	0.017 (0.018)
x11_process_innovation	0.235*** (0.063)	-0.020 (0.028)	0.236*** (0.063)	-0.022 (0.028)
x12_high_growth	-0.110+ (0.056)	0.008 (0.020)	-0.109+ (0.056)	0.009 (0.020)
x13_internationalization	-0.088 (0.054)	-0.080** (0.029)	-0.090 (0.054)	-0.082** (0.029)
x14_risk_capital	0.109* (0.045)	-0.035 (0.023)	0.109* (0.046)	-0.035 (0.023)
coherence × betweenness			-0.234*** (0.031)	-0.292*** (0.031)
coherence × eigenvector			-0.006 (0.030)	0.002 (0.030)
betweenness × rawdist				
rawdist × eigenvector				
Num.Obs.	1506300	1506300	1506300	1506300
R2	0.474	0.661	0.478	0.668
R2 Adj.	0.474	0.661	0.478	0.668
R2 Within		0.358		0.371
R2 Within Adj.		0.358		0.371
AIC	5748605.3	5088428.5	5735734.5	5057169.8
BIC	5748788.7	5092279.4	5735942.3	5061045.2
RMSE	1.63	1.31	1.62	1.30
Std.Errors	by: region & tech			
FE: region		X		X
FE: tech		Y		Y

appropriate given that the GEI variables are measured at the national level. The high intra-class correlation, $ICC = 0.60$, implies that 60 of variance in potential is between units rather than within-region over time. The variance components show that country-level heterogeneity (standard deviation approximately 1.12–1.17) exceeds region-within-country heterogeneity (approximately 1.02–1.04). These magnitudes do not contradict the fixed effects results; rather, they clarify that while the fixed effects estimates document robust within-region mechanisms, a substantial portion of the feasibility landscape is structured by relatively persistent differences across countries and regions.

Crucially, the core coefficients remain stable in the hierarchical models. In the random effects specification comparable to the fixed effects benchmark, coherence remains positive (around 0.709), betweenness remains positive (around 0.398), and the coherence $\times$ betweenness interaction remains negative (around -0.292). Eigenvector centrality remains negative (around -0.222) without coherence moderation. Distance is more salient in this setting: rawdist becomes significant and negative as a main effect (about -0.027), consistent with the idea that topological accessibility matters for cross-regional comparisons even if it is less informative for within-region shifts once fixed effects absorb time-invariant accessibility differences. At the same time, the interaction results reinforce the earlier asymmetry. The rawdist $\times$ eigenvector term remains strongly negative (around -0.082), indicating that remoteness compounds difficulty specifically for authority targets. The betweenness $\times$ rawdist interaction becomes positive and significant in the random effects model (e.g., 0.050–0.061), suggesting that bridging positions may be more relevant for overcoming remoteness in between-place comparisons than in within-region variation. Because

these patterns appear most clearly in the cross-sectional component captured by random effects, they should be interpreted as differences in accessibility across places rather than as short-run within-place responses.

These hierarchical results also help justify the spatial models that follow. First, the large *ICC* indicates that potential is strongly structured across places, not only through time-varying within-region dynamics. Second, (**TechNetworkResilience?**) emphasizes that regional economies are “webs” of interdependent activities and argues that “networks are of great assistance,” a framing that makes it plausible that feasibility is not independent across space but may be shaped by neighboring structures. Third, the distinction between within-region fixed effects patterns and between-place random effects patterns suggests that part of the remaining heterogeneity may be geographically patterned, motivating an explicit test for spatial dependence.

5.5 Role of space

Having established stable dyadic mechanisms at the technology–region level (Tables Table 5.2 and Table 5.3) and substantial between-place heterogeneity in the hierarchical models (Table Table 5.4), we test whether regional potential exhibits spatial dependence. We aggregate potential to region-year means and estimate spatial panel models with 5-nearest-neighbor weights matrices. Tables Table 5.5 and Table 5.6 reveal substantial spatial dependence. In the SAR model, the spatial autoregressive parameter is around 0.376 in the interaction specification, indicating that regional potential is positively associated

Table 5.4: logit(pred1) - ols

	inter_a	inter_al	inter_b	inter_b1	h_a	h_b
(Intercept)	-2.577*** (0.088)	-2.667*** (0.087)	-2.573*** (0.088)	-2.666*** (0.088)	-2.817*** (0.227)	-2.711*** (0.230)
patent_n	0.109*** (0.001)	0.108*** (0.001)	0.088*** (0.001)	0.088*** (0.001)	0.090*** (0.001)	0.090*** (0.001)
rawdist	-0.032*** (0.001)	0.008*** (0.002)	-0.029*** (0.001)	0.009*** (0.002)	-0.037*** (0.001)	-0.027*** (0.001)
betweenness	0.427*** (0.002)	0.200*** (0.001)	0.388*** (0.002)	0.147*** (0.001)	0.133*** (0.001)	0.398*** (0.002)
coherence	0.729*** (0.001)	0.802*** (0.001)	0.706*** (0.001)	0.780*** (0.001)	0.784*** (0.001)	0.709*** (0.001)
eigenvector	-0.212*** (0.001)	-0.223*** (0.001)	-0.221*** (0.001)	-0.233*** (0.001)	-0.240*** (0.001)	-0.222*** (0.001)
betweenness × coherence	-0.270*** (0.001)		-0.284*** (0.001)			
coherence × eigenvector	0.000 (0.001)		-0.001 (0.001)			0.002 (0.001)
betweenness × rawdist		0.061*** (0.004)		0.057*** (0.004)		
rawdist × eigenvector		-0.080*** (0.001)		-0.080*** (0.001)		
reg_comp			0.010*** (0.001)	0.014*** (0.001)	0.015*** (0.002)	0.011*** (0.002)
tech_comp			-0.254*** (0.001)	-0.255*** (0.001)	-0.243*** (0.001)	-0.241*** (0.001)
x8_human_capital					0.044*** (0.003)	0.044*** (0.003)
x9_competition					0.088*** (0.005)	0.088*** (0.005)
x10_product_innovation					0.016*** (0.002)	0.017*** (0.002)
x11_process_innovation					-0.020*** (0.004)	-0.022*** (0.004)
x12_high_growth					0.008** (0.003)	0.009** (0.003)
x13_internationalization					-0.080*** (0.003)	-0.082*** (0.003)
x14_risk_capital					-0.035*** (0.002)	-0.034*** (0.002)
coherence × betweenness						-0.292*** (0.002)
rawdist × betweenness						
SD (Observations)	1.346	1.356	1.307	1.318	1.310	1.296
SD (Intercept regioncountry)					1.029	1.036
SD (Intercept country)					1.124	1.135
SD (Intercept region)	1.497	1.489	1.513	1.499		
Num.Obs.	1999432	1999432	1906137	1906137	1506300	1506300
R2 Marg.	0.190	0.186	0.202	0.198	0.197	0.201
R2 Adj.	0.183	0.181	0.199	0.195	0.194	0.199

Table 5.5

	baseline	network	interaction
(Intercept)	0.116*** (0.003)	0.136*** (0.003)	0.132*** (0.003)
reg_comp	0.060*** (0.003)	0.055*** (0.003)	0.054*** (0.003)
patent_count	0.077*** (0.003)	0.062*** (0.003)	0.062*** (0.003)
rho	0.464*** (0.015)	0.375*** (0.016)	0.376*** (0.016)
avg_coherence		0.055*** (0.003)	0.065*** (0.003)
avg_betweenness		-0.009** (0.003)	0.009* (0.004)
avg_eigenvector		0.018*** (0.003)	0.022*** (0.003)
avg_coherence $\times$ avg_betweenness			0.013*** (0.002)
avg_coherence $\times$ avg_eigenvector			0.006*** (0.002)
Num.Obs.	3223	3223	3223
Log.Lik.	4350.51	4559.40	4578.55
AIC	-8693.02	-9104.81	-9139.09
BIC	-8668.71	-9062.26	-9084.39
Sigma	0.161	0.149	0.148
Res.Var	0.024	0.021	0.021
EDF Total	4	7	9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

with neighboring regions' outcomes. In the SEM model, the spatial error parameter is around 0.420 in the interaction specification, indicating spatial correlation in unobserved components. These estimates are consistent with the interpretation that spatial context matters for regional feasibility landscapes, whether through spillovers or through correlated shocks and unobservables.

The regional aggregate covariates also help interpret the relationship between dyadic feasibility and portfolio-level advantage. Average coherence is positive (about 0.065 in the

SAR interaction model and 0.073 in the SEM interaction model), consistent with the dyadic evidence that alignment is a central feasibility correlate. Average eigenvector centrality of existing portfolios is also positive (about 0.022 in SAR and 0.021 in SEM), and it interacts positively with average coherence (about 0.006 in SAR and 0.005 in SEM). This differs in sign from the dyadic eigenvector penalty for candidate technologies (around -0.22). A conservative reconciliation is that these statistics capture different objects: dyadic eigenvector describes the difficulty of entering an authority-like target technology, whereas average eigenvector of a region’s existing specializations reflects accumulated capabilities embedded in those technologies that may raise overall potential across the opportunity set. This distinction is compatible with the enabling-technology argument in (**GoingBeyondRelatednessKETs?**), which emphasizes that enabling capacities matter when they are embedded and used within the local system, rather than as isolated targets.

A similar distinction applies to betweenness. At the dyadic level, bridge positions are more valuable for low-coherence regions, reflected in the negative coherence $\times$ betweenness term (around -0.28). At the regional level, average betweenness is modest in isolation but becomes positive once interacted with average coherence (about 0.013 in SAR and 0.012 in SEM). This suggests that advantaged portfolios may combine alignment and bridging among what regions already do, while at the margin bridging targets provide relatively greater benefits when alignment is weak. Overall, the spatial results do not overturn the dyadic mechanisms; instead, they indicate that feasibility is also spatially structured, and that portfolio composition at the regional level can be associated with higher average

LISA Clusters – 2018

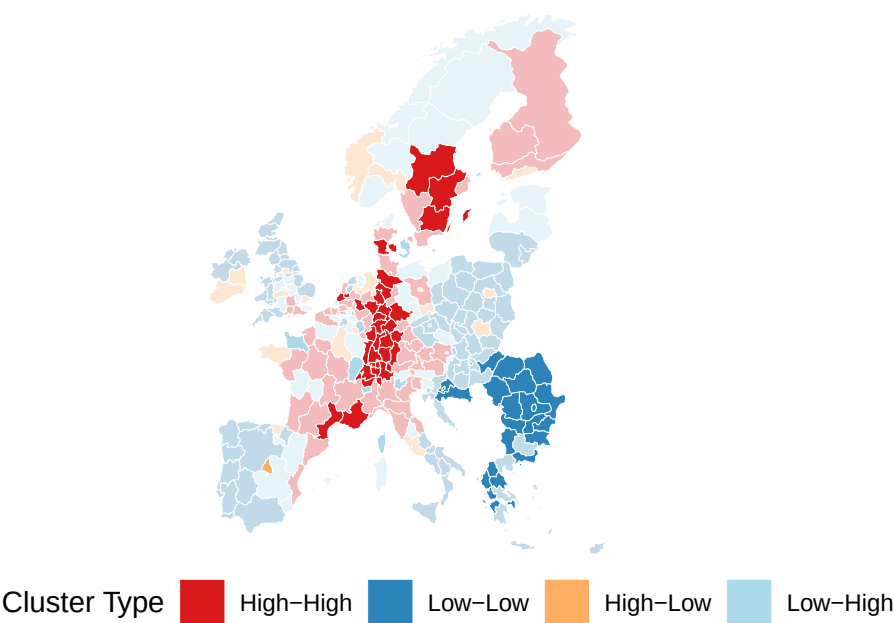


Figure 5.1

potential even when entry into specific authority targets remains difficult.

Table 5.6

	baseline	network	interaction
(Intercept)	0.217*** (0.006)	0.221*** (0.005)	0.217*** (0.005)
reg_comp	0.060*** (0.003)	0.059*** (0.003)	0.058*** (0.003)
patent_count	0.072*** (0.003)	0.059*** (0.003)	0.059*** (0.003)
delta	0.522*** (0.019)	0.421*** (0.023)	0.420*** (0.023)
avg_coherence		0.063*** (0.003)	0.073*** (0.004)
avg_betweenness		-0.008** (0.003)	0.009* (0.004)
avg_eigenvector		0.017*** (0.003)	0.021*** (0.003)
avg_coherence $\times$ avg_betweenness			0.012*** (0.003)
avg_coherence $\times$ avg_eigenvector			0.005** (0.002)
Num.Obs.	3223	3223	3223
Log.Lik.	4228.41	4443.00	4458.15
AIC	-8448.83	-8871.99	-8898.31
BIC	-8424.51	-8829.45	-8843.61
Sigma	0.181	0.162	0.161
Res.Var	0.025	0.023	0.022
EDF Total	4	7	9

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

6 Summary

6.1 Discussion

This work was initiated with a simple subjective reality check from geopolitical perspective. Since the financial crisis of 2008 the conventional rule-based world order started to corrode and languish. The covid 19 pandemic further exacerbated this corrosion and forced policy makers to reassess their decisions. This new world order is still in its infancy and its nature is yet to be clear, meanwhile uncertainties in global security and trade is only getting worse and vulnerable economies are the first victims. From here we posited that in the face of such shocks, understanding what makes an economy resilient is a sensible exercise. This was a clear research agenda in the literature which we presented and discussed in the review chapter. We framed adaptation as a theoretically observable consequence of resilience and that this adaptation, i.e. the capacity to adapt to constraints by recombining existing local capabilities in new configuration can be observed itself in the diversification patterns of these local economies. From there we justified the current study and framed it as a response to the need to a more contextualised assessment of regional technological

diversification. Our methodological and empirical contribution target essentially context dimension following the research agenda proposed in many studies in different streams of literature. In this chapter we will conclude this dissertation and discuss different aspect of novelty, limitations and implication to our contributions.

6.1.1 On the methodology

Our contribution in this work spans both methodology and empirical application. The call in the literature for a more contextualised studies of regional capabilities were our main theoretical motivation in this context. P.-A. Balland, Rigby, and Boschma (2015) call for a more focused consideration of the evolutionary accounts of regional change; technological relatedness, network structures, and institutions. However, C. Pinheiro (2025); Andreoni and Chang (2019) and (batheltstorper2023?) provided another optic regarding this agenda specifically on the limitation of the current conventional instruments of measuring proximity (relatedness) and thus the technology space (network) and their conceptual flaws that obscure granular context. This means that even if this research agenda is operationalised through relatedness measurements, the loss in information compound regardless. To address this specific issue, Fessina et al. (2024), Albora et al. (2023), and most notably Tacchella et al. (2023) propose a machine learning conceptualisation to the relatedness measurements. Their work propose a shift from co-occurrence and frequency based measurements to a predictive framework that serves three objectives. First, a falsifiable and testable measure of feasibility, second, leveraging already granular data for a richer context, and third accommodate the non-linearity of knowledge structure and their

interactions. We adopted this framework as a direct response to the gaps in the literature which constitutes our main methodological contribution. However, this methodology is not without limitation itself. In fact, C. Pinheiro (2025) acknowledged the merits of this framework with a clear factual scepticism regarding the performance of these machine learning models. The problem she mentions is related mainly to false negative predictions, which are the result of the unbalanced data itself (few regions with observed specialisation, the majority of the outcome is 0). Although this is a well merited criticism, it still can be addressed in future iterations with different training strategies (using synthetic minority oversampling technique or SMOTE) or with innovative machine learning modeling algorithms (causal random forest as an potential candidate). Moreover as we’ve shown in the comparative example in chapter 3, FITS produce a much more stable prediction than conventional technology space, the merits of that is our main motivation for this adoption.

The empirical design in our work is also not above criticism or limitations. Our asse

6.1.2 On the results

The main endogeneity concern in this analysis is that coherence is constructed from the same portfolio data that feeds the RF model generating potential. Within a given year and region, a technology with higher coherence will mechanically share portfolio features with the region in ways that likely raise the RF’s predicted probability. This is analogous to — though more subtle than — the reverse causality concern that Antonietti and Montresor (2021, p. 199, “Econometric Strategy”) address by controlling for firm birth

rates: regions endowed with more KETs may also be those where firm creation is normally higher, introducing spurious correlation between KETs and diversification outcomes. In our case, the circularity is internal to the variable construction rather than external to the model.

The two-way fixed effects address part of this: the identifying variation is within-region change over time, so stable portfolio-technology alignment is absorbed. The consistency of coefficients across fixed effects and random effects specifications, and the stability of the network centrality patterns — which are technology-specific rather than portfolio-derived — provide some reassurance that the results are not purely mechanical. We treat the estimates throughout as conditional associations that characterise the structure of the feasibility landscape rather than as causal parameters, and we return to the broader implications of this distinction in the discussion.

6.1.3 Policy implication

6.2 Future research agenda

6.3 Conclusion

References

- Acs, Zoltan J, Erkkö Autio, and László Szerb. 2014. “National Systems of Entrepreneurship: Measurement Issues and Policy Implications.” *Research Policy* 43 (3): 476–94.
- Acs, Zoltan J, Saul Estrin, Tomasz Mickiewicz, and László Szerb. 2018. “Entrepreneurship, Institutional Economics, and Economic Growth: An Ecosystem Perspective.” *Small Business Economics* 51 (2): 501–14.
- Albora, Giambattista, Luciano Pietronero, Andrea Tacchella, and Andrea Zaccaria. 2023. “Product Progression: A Machine Learning Approach to Forecasting Industrial Upgrading.” *Scientific Reports* 13 (1): 1481.
- Andreoni, Antonio, and Ha-Joon Chang. 2019. “The Political Economy of Industrial Policy: Structural Interdependencies, Policy Alignment and Conflict Management.” *Structural Change and Economic Dynamics* 48: 136–50.
- Antonietti, Roberto, and Sandro Montresor. 2021. “Going Beyond Relatedness: Regional Diversification Trajectories and Key Enabling Technologies (KETs) in Italian Regions.” *Economic Geography* 97 (2): 187–207.
- Argiles, Raquel Ortega, Zoltan J Acs, Laszlo Szerb, Erkkö Autio, and Eva Komlosi. 2014.

- “REDI: The Regional Entrepreneurship and Development Index—Measuring Regional Entrepreneurship Final Report.”
- Balassa, Bela. 1965. “Trade Liberalisation and ‘Revealed’ Comparative Advantage 1.” *The Manchester School* 33 (2): 99–123.
- Balland, PA, and R Boschma. 2019. “Smart Specialization: Beyond Patents.” *European Commission: Brussels, Belgium*.
- Balland, Pierre-Alexandre, and David Rigby. 2017. “The Geography of Complex Knowledge.” *Economic Geography* 93 (1): 1–23.
- Balland, Pierre-Alexandre, David Rigby, and Ron Boschma. 2015. “The Technological Resilience of US Cities.” *Cambridge Journal of Regions, Economy and Society* 8 (2): 167–84.
- Boschma, Ron. 2015. “Towards an Evolutionary Perspective on Regional Resilience.” *Regional Studies* 49 (5): 733–51.
- . 2017. “Relatedness as Driver of Regional Diversification: A Research Agenda.” *Regional Studies* 51 (3): 351–64.
- Boschma, Ron, Pierre-Alexandre Balland, and Dieter Franz Kogler. 2015. “Relatedness and Technological Change in Cities: The Rise and Fall of Technological Knowledge in US Metropolitan Areas from 1981 to 2010.” *Industrial and Corporate Change* 24 (1): 223–50.
- Boschma, Ron, and Simona Iammarino. 2009. “Related Variety, Trade Linkages, and Regional Growth in Italy.” *Economic Geography* 85 (3): 289–311.
- Boschma, Ron, Ernest Miguelez, Rosina Moreno, and Diego B Ocampo-Corrales. 2023. “The Role of Relatedness and Unrelatedness for the Geography of Technological Break-

- throughs in Europe.” *Economic Geography* 99 (2): 117–39.
- Bun, Joël, Jean-Philippe Bouchaud, and Marc Potters. 2017. “Cleaning Large Correlation Matrices: Tools from Random Matrix Theory.” *Physics Reports* 666: 1–109.
- Bustos, Sebastián, Charles Gomez, Ricardo Hausmann, and César A Hidalgo. 2012. “The Dynamics of Nestedness Predicts the Evolution of Industrial Ecosystems.” *PloS One* 7 (11): e49393.
- Christopherson, Susan, Jonathan Michie, and Peter Tyler. 2010. “Regional Resilience: Theoretical and Empirical Perspectives.” *Cambridge Journal of Regions, Economy and Society* 3 (1): 3–10.
- Coniglio, Nicola D, Davide Vurchio, Nicola Cantore, and Michele Clara. 2021. “On the Evolution of Comparative Advantage: Path-Dependent Versus Path-Defying Changes.” *Journal of International Economics* 133: 103522.
- Diodato, Dario, Lorenzo Napolitano, Emanuele Pugliese, and Andrea Tacchella. 2023. “Economic Complexity for Regional Industrial Strategies.” Joint Research Centre.
- E, Pugliese, and Tacchella A. 2021. “Economic Complexity Analytics: Country Fact-sheets.” <https://doi.org/10.2760/368138>.
- Fessina, Massimiliano, Giambattista Albora, Andrea Tacchella, and Andrea Zaccaria. 2024. “Identifying Key Products to Trigger New Exports: An Explainable Machine Learning Approach.” *Journal of Physics: Complexity* 5 (2): 025003.
- Fingleton, Bernard, Harry Garretsen, and Ron Martin. 2012. “Recessionary Shocks and Regional Employment: Evidence on the Resilience of UK Regions.” *Journal of Regional Science* 52 (1): 109–33.
- Frenken, Koen, and Ron A Boschma. 2007. “A Theoretical Framework for Evolution-

- ary Economic Geography: Industrial Dynamics and Urban Growth as a Branching Process.” *Journal of Economic Geography* 7 (5): 635–49.
- Hartmann, Dominik, Miguel R Guevara, Cristian Jara-Figueroa, Manuel Aristarán, and César A Hidalgo. 2017. “Linking Economic Complexity, Institutions, and Income Inequality.” *World Development* 93: 75–93.
- Hausmann, Ricardo, and César A Hidalgo. 2011. “The Network Structure of Economic Output.” *Journal of Economic Growth* 16: 309–42.
- Hekkert, Marko P, Roald AA Suurs, Simona O Negro, Stefan Kuhlmann, and Ruud EHM Smits. 2007. “Functions of Innovation Systems: A New Approach for Analysing Technological Change.” *Technological Forecasting and Social Change* 74 (4): 413–32.
- Hidalgo, César A. 2021. “Economic Complexity Theory and Applications.” *Nature Reviews Physics* 3 (2): 92–113.
- . 2023. “The Policy Implications of Economic Complexity.” *Research Policy* 52 (9): 104863.
- Hidalgo, César A, Pierre-Alexandre Balland, Ron Boschma, Mercedes Delgado, Maryann Feldman, Koen Frenken, Edward Glaeser, et al. 2018. “The Principle of Relatedness.” In *Unifying Themes in Complex Systems IX: Proceedings of the Ninth International Conference on Complex Systems 9*, 451–57. Springer.
- Hidalgo, César A, and Ricardo Hausmann. 2009. “The Building Blocks of Economic Complexity.” *Proceedings of the National Academy of Sciences* 106 (26): 10570–75.
- Hidalgo, César A, Bailey Klinger, A-L Barabási, and Ricardo Hausmann. 2007. “The Product Space Conditions the Development of Nations.” *Science* 317 (5837): 482–87.
- Komlósi, Éva, Marcus Dejardin, László Szerb, and Balázs Páger. 2025. “Is a Balanced

- Entrepreneurial Ecosystem Essential for Success? A Configurational Analysis of European Regional Entrepreneurial Ecosystems.” *The Journal of Technology Transfer* 50 (4): 1669–1708.
- Landau, William Michael. 2021. “The Targets r Package: A Dynamic Make-Like Function-Oriented Pipeline Toolkit for Reproducibility and High-Performance Computing.” *Journal of Open Source Software* 6 (57): 2959.
- Lee, Changyong, Ohjin Kwon, Myeongjung Kim, and Daeil Kwon. 2018. “Early Identification of Emerging Technologies: A Machine Learning Approach Using Multiple Patent Indicators.” *Technological Forecasting and Social Change* 127: 291–303.
- Leendertse, Jip, Mirella Schrijvers, and Erik Stam. 2022. “Measure Twice, Cut Once: Entrepreneurial Ecosystem Metrics.” *Research Policy* 51 (9): 104336.
- Li, Yang, and Frank MH Neffke. 2024. “Evaluating the Principle of Relatedness: Estimation, Drivers and Implications for Policy.” *Research Policy* 53 (3): 104952.
- Mariani, Manuel Sebastian, Zhuo-Ming Ren, Jordi Bascompte, and Claudio Juan Tessone. 2019. “Nestedness in Complex Networks: Observation, Emergence, and Implications.” *Physics Reports* 813: 1–90.
- Martin, Ron. 2012. “Regional Economic Resilience, Hysteresis and Recessionary Shocks.” *Journal of Economic Geography* 12 (1): 1–32.
- Martin, Ron, and Peter Sunley. 2007. “Complexity Thinking and Evolutionary Economic Geography.” *Journal of Economic Geography* 7 (5): 573–601.
- Mealy, Penny, and Diane Coyle. 2022. “To Them That Hath: Economic Complexity and Local Industrial Strategy in the UK.” *International Tax and Public Finance* 29 (2): 358–77.

- Montresor, Sandro, and Francesco Quatraro. 2017. "Regional Branching and Key Enabling Technologies: Evidence from European Patent Data." *Economic Geography* 93 (4): 367–96.
- Napolitano, Lorenzo, Evangelos Evangelou, Emanuele Pugliese, Paolo Zeppini, and Graham Room. 2018. "Technology Networks: The Autocatalytic Origins of Innovation." *Royal Society Open Science* 5 (6): 172445.
- Neffke, Frank, Matté Hartog, Ron Boschma, and Martin Henning. 2018. "Agents of Structural Change: The Role of Firms and Entrepreneurs in Regional Diversification." *Economic Geography* 94 (1): 23–48.
- Neffke, Frank, Martin Henning, and Ron Boschma. 2011. "How Do Regions Diversify over Time? Industry Relatedness and the Development of New Growth Paths in Regions." *Economic Geography* 87 (3): 237–65.
- Pendall, Rolf, Kathryn A Foster, and Margaret Cowell. 2010. "Resilience and Regions: Building Understanding of the Metaphor." *Cambridge Journal of Regions, Economy and Society* 3 (1): 71–84.
- Pike, Andy, Stuart Dawley, and John Tomaney. 2010. "Resilience, Adaptation and Adaptability." *Cambridge Journal of Regions, Economy and Society* 3 (1): 59–70.
- Pinheiro, Cristina. 2025. "Relatedness and Economic Complexity as Tools for Industrial Policy: Insights and Limitations." *Structural Change and Economic Dynamics* 72: 1–10.
- Pinheiro, Flavio L, Pierre-Alexandre Balland, Ron Boschma, and Dominik Hartmann. 2025. "The Dark Side of the Geography of Innovation: Relatedness, Complexity and Regional Inequality in Europe." *Regional Studies* 59 (1): 2106362.

- Pinheiro, Flávio L, Dominik Hartmann, Ron Boschma, and César A Hidalgo. 2022. “The Time and Frequency of Unrelated Diversification.” *Research Policy* 51 (8): 104323.
- Rocchetta, Silvia, Andrea Mina, Changjun Lee, and Dieter F Kogler. 2022. “Technological Knowledge Spaces and the Resilience of European Regions.” *Journal of Economic Geography* 22 (1): 27–51.
- Saracco, Fabio, Riccardo Di Clemente, Andrea Gabrielli, and Tiziano Squartini. 2015. “Randomizing Bipartite Networks: The Case of the World Trade Web.” *Scientific Reports* 5 (1): 10595.
- Saracco, Fabio, Mika J Straka, Riccardo Di Clemente, Andrea Gabrielli, Guido Caldarelli, and Tiziano Squartini. 2017. “Inferring Monopartite Projections of Bipartite Networks: An Entropy-Based Approach.” *New Journal of Physics* 19 (5): 053022.
- Simmie, James, and Ron Martin. 2010. “The Economic Resilience of Regions: Towards an Evolutionary Approach.” *Cambridge Journal of Regions, Economy and Society* 3 (1): 27–43.
- Szerb, Laszlo, Zoltán Ács, Erkkó Autio, Raquel Ortega-Argiles, and Eva Komlosi. 2013. “The Regional Entrepreneurship and Development Index—Measuring Regional Entrepreneurship.”
- Tacchella, Andrea, Andrea Zaccaria, Marco Miccheli, and Luciano Pietronero. 2023. “Relatedness in the Era of Machine Learning.” *Chaos, Solitons & Fractals* 176: 114071.
- Tóth, Gergő, Zoltán Elekes, Adam Whittle, Changjun Lee, and Dieter F Kogler. 2022. “Technology Network Structure Conditions the Economic Resilience of Regions.” *Economic Geography* 98 (4): 355–78.
- Whittle, Adam, and Dieter F Kogler. 2020. “Related to What? Reviewing the Literature

- on Technological Relatedness: Where We Are Now and Where Can We Go?” *Papers in Regional Science* 99 (1): 97–114.
- Wright, Marvin N, and Andreas Ziegler. 2017. “Ranger: A Fast Implementation of Random Forests for High Dimensional Data in c++ and r.” *Journal of Statistical Software* 77: 1–17.
- Xiao, Jing, Ron Boschma, and Martin Andersson. 2018. “Resilience in the European Union: The Effect of the 2008 Crisis on the Ability of Regions in Europe to Develop New Industrial Specializations.” *Industrial and Corporate Change* 27 (1): 15–47.
- Zaccaria, Andrea, Saurabh Mishra, Masud Z Cader, and Luciano Pietronero. 2018. “Integrating Services in the Economic Fitness Approach.” *World Bank Policy Research Working Paper*, no. 8485.
- Zhu, Shengjun, Qi Guo, and Canfei He. 2021. “Strong Links and Weak Links: How Do Unrelated Industries Survive in an Unfriendly Environment?” *Economic Geography* 97 (1): 66–88.